

PAMP's Exposure Increases the Areas of Pathological Tau and Inflammation in the PS19

Mouse Model of Alzheimer's Disease

Joshua Keene

Central Michigan University

Doctor of Philosophy in Neuroscience

Journal Article

May 2024

Acknowledgments

I would like to acknowledge several individuals and institutions that made this dissertation possible. I am extremely grateful to you all for your guidance, hard work, advice and support throughout this experiment. I would like to thank my advisors at Central Michigan University, including Dr. Gary Dunbar, Dr. Julien Rossignol, and Dr. Bhairavi Srinageshwar. I would also like to thank my fellow colleagues, including Swathi Suresh. I would like to thank Central Michigan University, including the Department of Psychology and the Neuroscience Program faculty. I would also like to thank the Field Neuroscience Institution Lab for Restorative Neurology.

I would like to thank my advisors and mentors at Michigan State University Grand Rapids Research Center, including Dr. Dave Morgan and Dr. Marcia Gordon. I would also like to thank my colleagues at MSU GRRC, including Taylor Desjarlais, Brianna Jackman, and Dr. Dylan Finneran. I would also like to thank the Department of Translational Neuroscience, Michigan State University, Grand Rapids, MI. Without the above-mentioned individuals and resources, this experiment would not have been possible.

Abstract

Alzheimer's disease (AD) is the most predominant form of dementia, marked by severe cognitive decline and eventually death. One hallmark of AD is the accumulation of pathological tau protein. This study used the PS19 mouse model harboring humanized P301S microtubule-associated tau protein transgenes to study the effects of pathogen-associated molecular patterns (PAMPs) on tau deposition and neuroinflammation. Four different PAMPs, each affecting a different toll-like receptor, were administered intraperitoneally one time per month starting at 6 months of age. Behavioral results showed that PS19+ animals that received PAMPs had significantly reduced basic movements and rearing behaviors in the open-field test. Immunohistochemical analysis demonstrated that PAMP administration significantly increased pathological tau accumulation in both whole-brain and hippocampal tissue, as well as reactive astrocytes in whole-brain tissue. Major histocompatibility complex II on microglia were significantly elevated in both whole-brain and hippocampal regions of mice receiving PAMPs. These findings suggest that PAMPs increase the pro-inflammatory environment found in AD, as well as the rate of pathological tau deposition.

Keywords: Alzheimer's disease, inflammation, tau, pathogen associated molecular patterns, microglia, astrocytes.

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Introduction

Alzheimer's disease (AD) is a prevalent neurodegenerative disease that results in severe cognitive decline and behavioral deficits. Currently, no effective options for preventing or treating AD are available. AD afflicts an estimated 6.7 million individuals in the United States (Alzheimer's Association, 2023). It is estimated that 79.8 million people worldwide will be diagnosed with AD by the year 2050 (Brookmeyer, 2018). Multiple variables likely contribute to the cause and/or progression of this disease.

Some of the hallmarks of AD pathophysiology include amyloid- β aggregation, ultimately leading to A β plaque deposition (He, 2018), tau dissociation from microtubules, which leads to the formation of neurofibrillary tangles (NFTs) (Iqbal, 2016) and heightened chronic neuroinflammation (Heneka, 2015). A β plaques, tau accumulation and neuroinflammation precede synaptic dysfunction, whereby existing connections degenerate and lose their integrity (Jackson, 2019) leading to a progressive increase in neuronal cell death (Jacobs, 2012).

Tau is a stabilizing protein that becomes compromised, contributing to the onset and progression of AD. The microtubule associated protein tau (MAPT) gene codes for the tau protein. Tau normally functions as a stabilizing protein that adheres to microtubules that are primarily found in neurons. Tau is a small (36-46 kDa) microtubule associated protein, encoded by the microtubule-associated protein tau (MAPT) gene. Under normal physiological conditions, tau binds to, and stabilizes, microtubules in neurons which transport a wide array of cellular components and supports axonal outgrowth and synaptic stability. Humans possess 3R and 4R isoforms, while mice possess 4R isoforms. In AD, the increased ratio of 4R compared to 3R isoforms correlates with pathology observed in human post-mortem brain analyses (Lacovich,

2017). Therefore, the balance of 3R to 4R tau isoforms may be an important factor in the development of AD. The PS19 mouse model contains the P301S human mutation, which only produces mutations in the 4R tau isoform (Zareba-Parlawska, 2020). Some 3R gene isoforms will be expressed to a certain degree, because the humanized mutation that the PS19 mice receive can be alternatively spliced out during gene expression, including the point mutation present in carrier animals of these models. Both isoforms of tau are integral parts of the cellular architecture in humans, including ubiquitous localization in axon compartments of neurons. In humans afflicted with AD, evidence suggests that tau dissociation contributes to many aspects of the AD phenotype (Khan, 2016).

Tau hyperphosphorylation causes microtubule dissociation. Eventually, phosphorylated tau (ptau) becomes soluble and begins to aggregate within the axonal compartment of neurons, preceding loss of synaptic structure and function (Swanson, 2017). Formation of soluble tau protein progressively aggregates, forming toxic tau paired helical filaments (PHFs) and subsequent NFTs. Several variables contribute to hyperphosphorylation of tau, including the presence of abnormally high amounts of A β monomers (Zempel, 2010) and A β fibril accumulation (Busciglio, 1995). The phagocytic capacity of microglia and astrocytes during periods of chronic inflammation progressively causes dissociation of MAPT and aggregation of tau protein (Trinczek, 1995). Tau phosphorylation inordinately affects axonal microtubule support, which in turn, causes structural destabilization of the cell and impairment of transportation of a wide array of critical cellular components (Ballatore, 2016). Tau dissociation, misfolding, and aggregation also correlates with synaptic dysfunction (Li, 2018). This

phenomenon precedes neuronal death and cognitive impairments associated with AD (Bejanin, 2017).

Inflammation contributes to AD progression and may also be an underlying mechanism that causes AD. Humans with AD typically display a pro-inflammatory state, marked by gliosis and increased production of pro-inflammatory cytokines. Pro-inflammatory cytokines are present in elevated levels in AD, including, interleukin-1 β (IL- β), IL-1 α , IL-6, and tumor necrosis factor alpha (TNF- α) (Rubio-Perez, 2012). The presence of pathogens and immune-system-mediating proteins in the brain of individuals with AD, suggests that exposure to pathogens and a heightened immune response contribute to AD.

Microglia are responsible for much of the detection and clearance of infiltrating peripheral material or damage due to brain injury. Evenly distributed throughout the brain under normal circumstances, microglia are the primary immune cells of the CNS. They form extensions called filipodia, to survey the extracellular space. These filipodia detect abnormalities in the extracellular fluid, including changes in ion concentrations that indicate abnormal structural and functional changes have taken place. The quantity and morphology of microglia vary, depending on the local environment. Resting microglia contain long ramified processes and a small soma. Reactive microglia have bushy filipodia and moderately sized soma. Ameboid and phagocytic microglia have few processes, large somas, and are ready to engulf debris. During microgliosis, the quantity of microglia increases in response to perceived pathogens, the presence of debris, or brain insult (Moore, 2019). Disease-associated microglia undergo a pro-inflammatory phenotypic shift, with an associated loss of homeostatic function (Wang, 2015). The phagocytic ability of microglia declines during the progression of AD, as A β

plaque deposition increases, tau-associated tangles become more numerous, and when synapses lose their integrity and are prematurely degraded and pruned (Nizami, 2019).

Astrocyte immune-related functions are prominent and diverse. Like microglia, astrocyte morphology varies depending on their environment. Astrocytes perform a critical role maintaining the integrity of the BBB, including staving off infiltrating pathogens and peripheral materials (Zenaro, 2017). Astrocytes also transfer glucose through end-feet located adjacent to blood vessels, providing the brain with a continual supply of glucose, which is disproportionately needed to maintain cellular function. In AD, astrocytes become reactive, leading to a reduction in glucose utilization by neurons (Steele, 2012). Astrocytes also modulate synaptic transmission of ions and neurotransmitters between neurons, and this process is degraded during AD, leading to synaptic loss (Hulshof, 2022). The pro-inflammatory state that is typical in AD, is to a large extent produced by astrocyte activity, in concert with microglia. Like microglia, astrocytes help to shift the immune response, depending on the state of their local environment. Astrocytes and microglia are capable of altering the immune response in their local environment through cytokine production and proliferation (Singh, 2022). Astrocytes are prone to influence from cytokines produced by themselves (as an autocrine function), or by neighboring cells (as a paracrine function). Upon exposure to lipopolysaccharides (LPS), astrocytes use toll-like receptors (TLRs) 2 and 4. TLR 2 and 4 activation subsequently activates a pro-inflammatory cellular cascade, upregulating production of TNF- α . TNF- α is a necessary factor for the proliferation of astrocytes during gliosis (Rogers, 2008). Loss of blood brain barrier (BBB) integrity through activation of reactive astrocytes and other methods leaves the CNS vulnerable to infiltration of cytokines and microbes from the periphery.

Peripheral proinflammatory cytokines, such as IL-1 β , IL-6, IL-18, and TNF- α , are significantly increased in humans with AD (Park, 2020). Sepsis has been shown to increase the rate of AD-related pathological hallmarks, including extracellular A β and intracellular tau deposition (Wang, 2018). Evidence suggests that certain infectious diseases contribute to the onset and progression of AD pathology (Baudron, 2015). Human herpes virus and hepatitis C are both causally linked to the onset of AD (Sochocka, 2017). Sochocka and colleagues correlated increased progression of the AD phenotype with bacterial infections, fungi, and protozoal infections. Microbial infiltration of the BBB, as well as indirect stimulation of microglia and astrocytes, exacerbate the conditions found in AD. Furthermore, activation of the peripheral immune system in response to pathogens increases the probability of peripheral immune cell infiltration into the brain. When the BBB is compromised, circulating leukocytes, including neutrophils, macrophages and T-cells, can become overly invasive and eventually deleterious to the brain in AD (Rossi, 2011). Inflammation is at the center of many aspects of AD, however, the underlying mechanisms and temporal relevancy of the inflammatory response taking place in AD is complex and still poorly understood.

TLRs are membrane-bound receptors responsible for inducing a pro-inflammatory state. Approximately 10 active TLR varieties can be found in humans, while at least 12 are found in mice. Neurons, microglia, and astrocytes are densely rich in TLRs. Pathogen associated molecular patterns (PAMPs) are specific to pathogens, such as viruses, bacteria, and parasites. Damage-associated-molecular patterns (DAMPs) are derived from debris from damaged cells of the host. TLRs are one family or version of pattern-recognition receptors (PPRs), they enable the recognition of ligands (proteins) containing common patterns, such as motifs associated with

perceived pathogens (PAMPs) or debris (DAMPs) (Kawai, 2010). These microbe-and-injury-related ligands bind to TLRs and cause downstream effects that signal the cell to coordinate a response to alterations in the normal environment.

Ligand-bound TLRs trigger innate immune responses, ultimately signaling transcription factors that induce a diverse immune response, including stimulation of nuclear factor kappa-light-chain-enhancer of activated B cells (NF- κ B) and interferon regulatory factors (IRF). NF- κ B-mediated gene expression is spatially selective (dependent on the site of TLR activation), and also particular to the type of pathogen present (also called event-related activation). While NF- κ B is well known for its influence over pro-inflammatory cytokines, such as IL-1 β , IL-6, and TNF- α , it has many other inflammation-related roles. NF- κ B affects B cell production, migration, and adhesion (Hayden, 2011). Leukocyte adhesion molecules are mediated through NF- κ B, such as E-selectin, vascular cell adhesion molecules (VCAM-1), and intercellular adhesion molecules (ICAM-1). Cyclooxygenase protein expression is increased by NF- κ B. IRF transcription factor activity is pathogen-dependent and location-based. IRFs respond to viral presence. The production of T helper cells is one of the primary influences of IRF (Brustle, 2007). Interferon regulatory factor activation induces innate immune system pattern recognition receptors, while simultaneously being activated by the adaptive immune system to promote viral recognition and anti-viral defenses (Yanai, 2012). Recognition of highly conserved antigens on pathogens enables TLRs to recognize and respond to a variety of perceived threats, ultimately facilitating a robust and diverse immune response.

This study focuses on four different types of PAMPs, each activating a specific TLR. Zymosan A activates TLR-2. This is a glucan, derived from the cell wall of the yeast,

Saccharomyces cerevisiae. Administration of Zymosan A stimulates sepsis, shock, and peritonitis. Polyinosinic-Polycytidylic acid (Poly I:C) activates TLR-3. This is a synthetic analog of viral dsRNA. Administration of Poly I:C triggers the production of the pro-inflammatory cytokines TNF- α , IL-6, and CXCL10. Cytosine phosphorothioate-guanine (CpG) oligonucleotide (ODN 2395) activates TLR-9. Class C CpG mimics bacteria/viral infection. Administration results in IFN- α production (from B-cells), which activates plasmacytoid dendritic cells (pDC). LPS primarily binds to and activates TLR-4. Administration of glycolipids derived from gram-negative bacterial cell wall (endotoxin) results in a pro-inflammatory state, and potentially produces septic shock. We hypothesize that PAMPs increase neuroinflammation and pathological tau accumulation in the PS19 mouse model of AD.

Materials and Methods

Animals and Experimental Design

Eighty-two PS19 and C57BL/6-C3H/HeJ non-transgenic (Non-Tg) littermates were used in this study (Jackson Laboratories). These animals were balanced for sex (figure 1). Twelve-week-old male breeders, hemizygous for the humanized MAPT transgene (P301S), were paired with two 12-week-old Non-Tg(noncarrier) female littermates, producing filial 1 (F1) generation male and female mice. PS19 hemizygous mice harbor the Tg(Prnp-MAPTP301S)PS19Vle/J gene mutation. This MAPT gene is under the control of the prion protein promoter (Prnp), which begins expression as early as embryonic day 12.5 (Asante, et al., 2002). Hemizygous PS19 positive mice exhibit tau microtubule dissociation and tau protein aggregation compared to their noncarrier littermates. This tauopathy model of AD lacks a significant APP-related disease phenotype compared to other mouse models typically harboring APP/PSEN mutations that

cause APP dysfunction and protein aggregation. The PS19 mouse model induces several pathological hallmarks consistent with AD, most notably tau protein pathology (figure 2).

The mice were housed with matching sex littermates in polyethylene cages with wood chip bedding, crinkled paper, and a half sheet of paper towel. This paper is primarily used for hiding and nesting behavior, reducing the probability of aggressive behavior between littermates, as well as improving thermoregulation, and finally, providing an environmental enrichment source. No more than 5 mice were housed in a single cage. All mice were given access to food pellets and water *ad libitum* and kept under 12h light-dark cycles (lights on at 0900 hours) at a constant 20° C temperature. All procedures using these mice were approved by the Institution of Animal Care and Use Committee (IACUC) at Central Michigan University. Mouse breeder cages included cardboard refuge huts for comfortable nesting and litter-care, as well as material to shred.

Intraperitoneal injections were given to mice one time every month starting at 6 months of age. Mice were given either all PAMPs or all Ringers injections one time per month until 9 months of age, with a total of 4 injections per mouse. The order of injections was kept the same for all PAMPs-injected mice, which consisted of Zymosan A, Poly I:C, ODN 2395 and LPS (figure 3). A 30-gauge 5/16-inch length needle was used with an insulin syringe for these injections. Each PAMP was measured and diluted into Ringers Mammalian solution to achieve the desired concentration. This solution was then mixed gently using a vortex mixer. This method ensured that each dose was evenly diffused in the working solution before administration. Solutions were kept cold (on ice) until shortly before use. Solutions were allowed to reach room

temperature before injections, in order to avoid causing hypothermia. The volume of solution administered was based on mouse weight (per gram).

PS19 PAMP Study Design

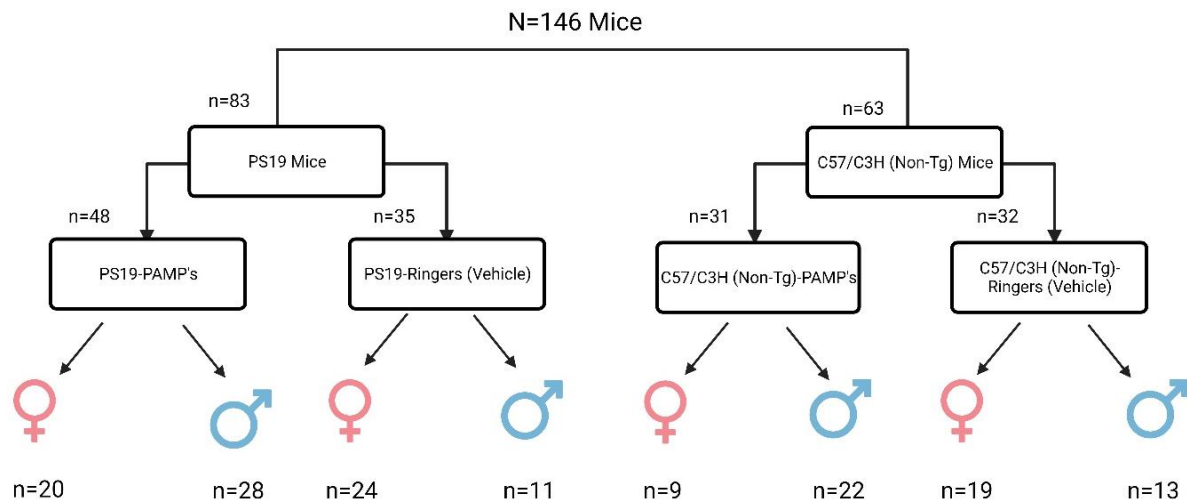


Figure 1: Overall study design demonstrating all 4 groups, including; PS19+ mice that received PAMPs, PS19+ mice that received Ringers, Non-Tg mice that received PAMPs, and Non-Tg mice that received Ringers. This study attempted to balance both sexes in and between each group.

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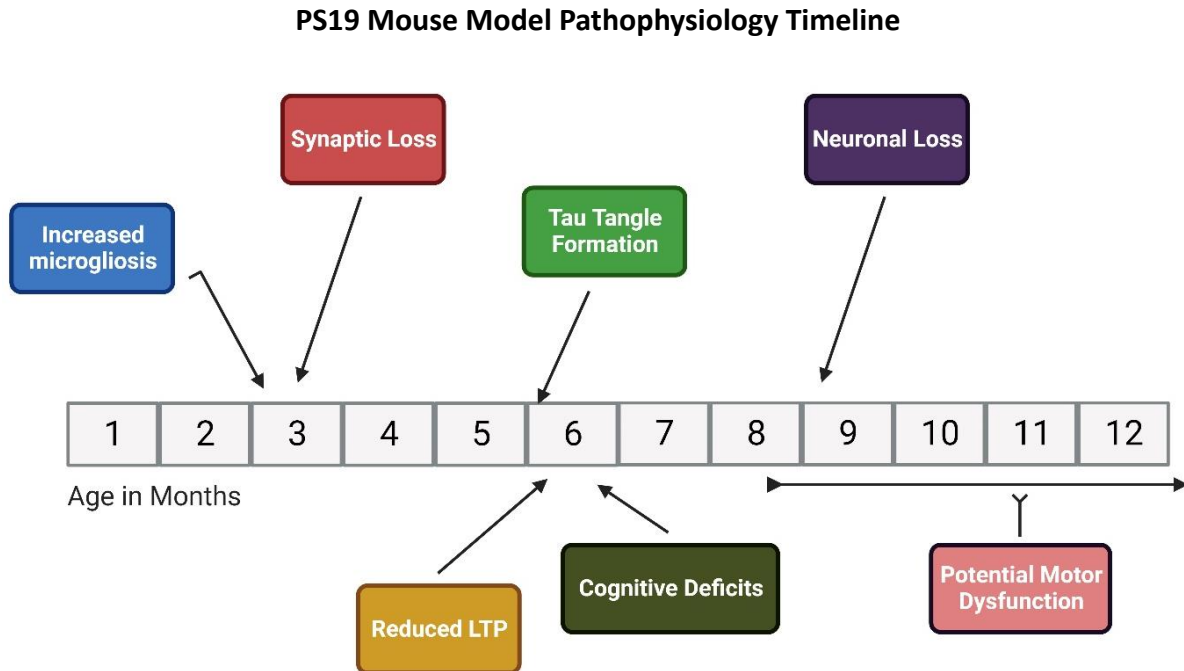


Figure 2: Pathophysiological development during the lifespan of P301S positive mice.

Microgliosis begins early, around 3 mo. Tau tangle formation begins around 6 mo. Motor dysfunction begins to afflict some mice harboring the P301S MAPT transgene starting close to 9 mo.—while the onset and severity of motor dysfunction varies between mice, most are able to function past 10 mo. Redrawn from <https://www.alzforum.org/research-models/tau-p301s-line-ps19>.

PAMP Concentrations and Administration

Zymosan A was mixed into a working concentration of 50 mg/kg. Poly I:C was mixed into a working solution of 12 mg/kg. ODN2395 was diluted into a working solution of limulus amoebocyte lysate (LAL) endotoxin free water at a concentration of 2.5 mg/kg.

Lipopolysaccharides (LPS) derived from *Escherichia coli* were the last PAMP administered, at a working solution of 1 mg/kg. LPS was sourced and diluted from a single batch for all mice receiving PAMPs, in order to avoid inter-batch variability common in LPS products. Glass vials,

rather than polyethylene containers, were used because glass significantly reduces adherence of LPS to the wall of the containers, providing a reliable and valid dose for each animal.

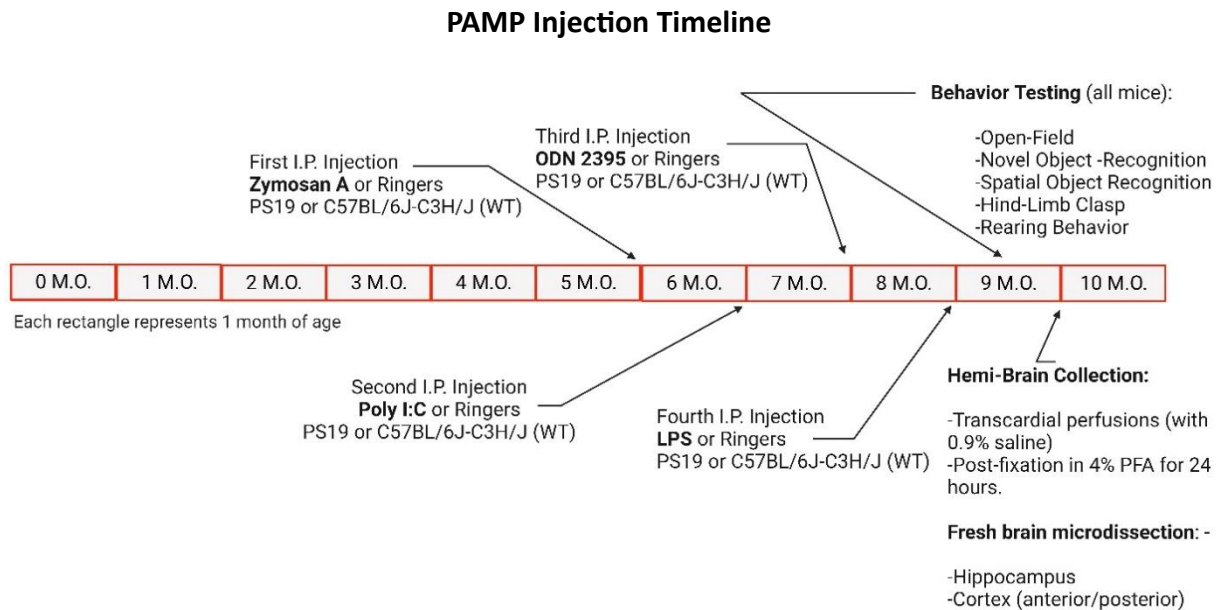


Figure 3: Timeline for I.P. injections of PAMPs or vehicle to PS19+ or Non-Tg mice. This timeline also includes behavioral testing, starting at 9 ½ months of age, and hemi-brain collections at 10 months old.

Behavioral Assessments

Each mouse underwent behavioral testing starting at 9 ½ months of age, prior to euthanasia and tissue collection at 10 months of age. All behavioral tests were performed blind to the genotype and treatment type. Each behavioral test was conducted on separate, but consecutive days, in the same order for each mouse: (1)Hind-Limb Clasp test (2) Open-Field behavioral test. Overhead lighting was evenly distributed, at 70-72 lux. Each mouse cage was placed in the behavior room for 1 hour prior to the onset of behavioral testing, in order to habituate each mouse to their environment prior to testing. Mice were tested starting at 0900 hours.

Hind-limb clasp behavioral testing consisted of gently picking up the mouse by its tail and holding it up for 10 seconds while video-recording the movements (figure 4A-B). Hind-Limb Clasp behavioral testing and scoring was conducted using previously described guidelines (Guyenet, 2010). Analysis was performed by scoring the duration of time each mouse spends displaying clasping behavior, consisting of contracting their hind limbs towards their abdomen inwards (figure 4B). Typical posture in healthy mice consists of splaying behavior observed in mice with normal motor function—outward stretched hind-limb posture (figure 4A). This measure was simplified into a categorical score of 0-3. A score of 0 is recorded when a mouse displays consistent (normal) splaying behavior, with their hind-limbs stretched outwards, away from their abdomen. A score of 1 was recorded when a mouse retracted one hind-limb towards its abdomen. When both hind-limbs are partially retracted towards the abdomen, then a score of 2 was provided. Mice that fully retracted their hind-limbs touching their abdomen for more than 50% of the time were given a score of 3.

Open-field behavioral testing was assessed using HK MotorMonitor software (Kinder Scientific, San Diego, CA) conducted in a 63 long x 36 wide x 33 high (in cm) polyethylene enclosure with laser grids used to quantify different aspects of mouse behavior (figure 4B). Mice were placed into a large rectangular polyethylene cage container. Each mouse spent 10 minutes in the open container during this test. HK MotorMonitor software measured four different variables: (1) Basic movement), (2) fine motor movement, (3) total distance (total exploratory movements), and (4) rearing behavior.

Hind-Limb Clasp and Open-Field Behavioral Testing

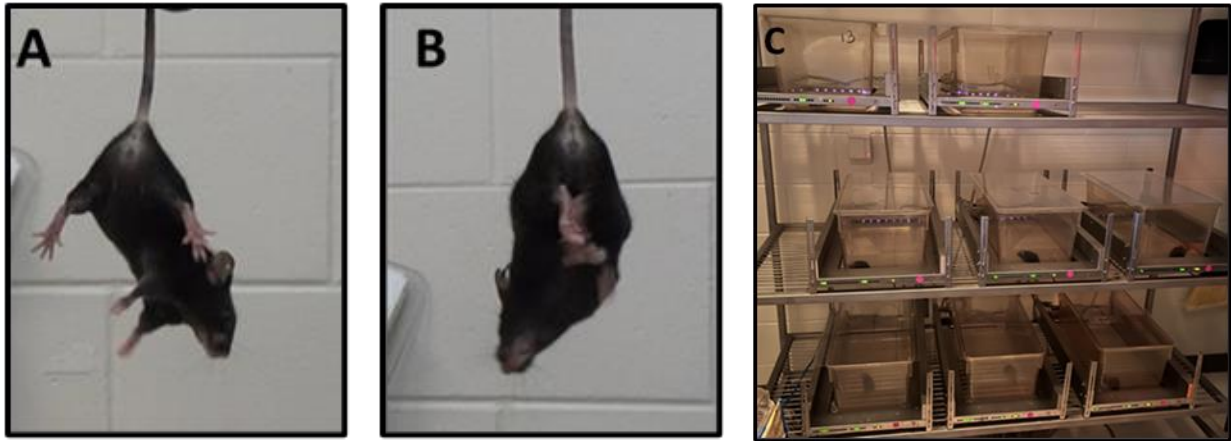


Figure 4: Behavioral testing was conducted for this study starting at 9 ½ M.O., before tissue collection. The Hind-Limb Clasp behavioral test was conducted; **A.** Depicts a Non-Tg mouse demonstrating typical splaying behavior of the hind-limbs, with a score of 0. **B.** Depicts a PS19+ mouse displaying clasp (retracted) hind-limb posture with a score of 3. The Open-Field behavioral test was also conducted; **C.** Represents the open-field behavioral testing apparatus (HK MotorMonitor) using large rat cages and a laser grid system to quantify different aspects of movement, including: total distance, basic movement, fine motor movement and rearing behavior.

Tissue Collection

Following behavioral testing, the mice were euthanized using an overdose of pentobarbital I.P. injections. Transcardial perfusions were used to clear blood from tissue before each brain was collected. Ice-cold 0.9% saline (made fresh from 10x stock solution) with 4.7 mg/L of heparin (anticoagulant), was delivered into the atrium at a flow rate of 7 mL per minute, with a total volume of 35 ml per mouse. After each brain was extracted, one hemisphere was immediately immersed in freshly made 4% paraformaldehyde (pH 7.2) for 24

hours and stored at 4° C. After this 24-hour incubation period, the tissue was changed out and incubated in PBS with a 10 mM concentration of sodium azide, in a sucrose gradient of 10%, 20%, and 30%, for 24-hours each. The sucrose gradient was used to prevent the formation of ice crystals during cryopreservation by displacing water with ever-increasing concentrations of sucrose, until the tissue sinks to the bottom of the container. For long-term storage, tissue was incubated in cryoprotectant media and stored in a 4° Celsius fridge until each brain was sectioned, as previously reported (Morgan, 2001).

The remaining half of the brain was microdissected on a metal tray resting flat on a cold stage, immediately after the saline perfusion. Fresh tissue was separated into hippocampus, midbrain, cerebellum and rest of the tissue (whole-brain samples) in polyethylene tubes. All tissue was flash frozen after being immediately immersed in ice cold methyl butane (at -80 C), and then wrapped in aluminum foil and temporarily stored on dry ice. After the completion of collections, the frozen tissue was stored at -80° C for long-term storage.

Histopathology and Microscopy

Mouse hemibrains were removed from sucrose solution and frozen on a sliding microtome stage using O.C.T. medium (Fisher Scientific, cat. 23-730-571) and sectioned in the horizontal plane at 25 µm thickness. Tissue sections were placed into free-floating staining trays for each immunohistochemical (IHC) marker, with multiple tissue sections in each well. Tissue sections were selected from serial sections, for a total of 8 tissue sections per stain, approximately every 400 µm between each section.

IHC was conducted according to the methods outlined previously (Gordon, 2002). It should be noted that for each separate antibody, mice of all genotypes, treatment and sex were

stained using the same antibody solution and staining trays, to avoid the possibility of differing concentrations or differences in antibody batches—increasing the uniformity and reliability of each antibody reaction. It should also be noted that not all mice were stained using all methods (see the results section for individual IHC sample sizes of each group based on the antibody used). All tissue was incubated in blocking solution for 15 minutes at room temperature (RT) on a shaker at 40 rotations per minute (RPM). Blocking solution was used to eliminate endogenous peroxidases (10% methanol, 3% H₂O₂ in phosphate buffered saline (PBS) pH 7.4). Tissue was then rinsed three times with PBS. Afterwards, the tissue was incubated in permeabilization solution (0.2% lysine, 1% Triton x 100, 4% normal goat serum (NGS) in PBS) for 30 minutes at RT at 40 RPM. The tissue was then incubated in the primary antibody overnight (12 hours). The primary antibodies used in this study include Phospho-Tau (Ser202, Thr205) (AT8), Invitrogen MN1020B at 1:1k, glial fibrillary acidic protein (GFAP), Agilent Z033401-2 at 1:50k, major histocompatibility complex two (MHC II), Invitrogen 14-5321-82 at 1:1k, and cluster of differentiation 68 (CD68), BioRad MCA1957 at 1:1k concentration. After the primary antibody incubation, the tissue was rinsed three times with PBS and then incubated in the corresponding secondary antibody at a concentration of 1:3k for two hours at RT at 40 RPM. The secondary antibodies used include goat-anti-mouse, Abcam, ab6788, goat-anti-rat, Invitrogen, 31830, and goat-anti-rabbit, Vector, BA-1000. The tissue was then rinsed three times in PBS and incubated for one hour at RT at 40 RPM in Vectastain Elite ABC kit solution (Vector Laboratories, PK-6100). After this one-hour incubation period, the tissue was then rinsed two times in PBS and a third rinse with tris buffered saline pH 7.4 (TBS). The tissue was then incubated in 3,3'-diaminobenzidine (DAB) substrate for chromogenic color development (0.05%

diaminobenzidine, 0.5% Ni²⁺, 0.03% H₂O₂) in pH 7.4 TBS until color developed. All tissues for each antibody were incubated in DAB for the same period of time, but different antibodies required different DAB incubation periods, with a maximum time-period of 5 minutes. It should be noted that not all of the IHC markers included Ni²⁺ amplification. DAB enzymatic reactions were stopped by removing tissue and rinsing it in TBS, and then two subsequent rinses in PBS. Once all of the staining was complete, the tissue was mounted on positively-charged glass microscope slides (Fisher Scientific, 1255015). The tissue was then dehydrated, and cover slipped with DPX mounting medium (Electron Microscopy Science, 13512).

Slides were digitally scanned using a 20x objective (200 x's magnification) using a ZEISS Axio Scan.Z1 microscope, and then subsequently analyzed using NearCYTE IHC analysis software (NearCYTE, Pittsburgh, PA). NearCYTE software was initially used to establish the regions of interest for each section. This included whole-brain regions and hippocampus regions. NearCYTE software was then used to establish a threshold for immunopositive pixels. Thresholds were identified for each separate marker and were uniform within each biomarker being observed and quantified. Thresholds were selected for the best signal to noise ratio and to reduce the inclusion of non-specific staining. Multiple random samples for each antibody were observed to determine the threshold for each marker. Calculations for each tissue section were based on the ratio of immunopositive pixels to the total pixel area for each tissue section—which provided us with a % of area stained for each IHC group.

Results

Behavioral Assessments

Unpaired parametric T-tests were used to determine if treatment or genotype influenced splaying behavior using the hind-limb behavioral test. PS19+ mice that received PAMPs showed significantly elevated splaying behavior ($p=.0051$), with a composite mean score of 1.615 ($n=13$) compared to Non-Tg mice that received PAMPs, which showed a mean score of 0.3571 ($n=14$). There was also a significant difference based on genotype between PS19+ mice and Non-Tg mice that received only Ringers ($p=.0408$). PS19+ mice that received Ringers had a mean score of 1.077 ($n=13$), while Non-Tg mice that received Ringers had a mean composite score of .3750 ($n=16$).

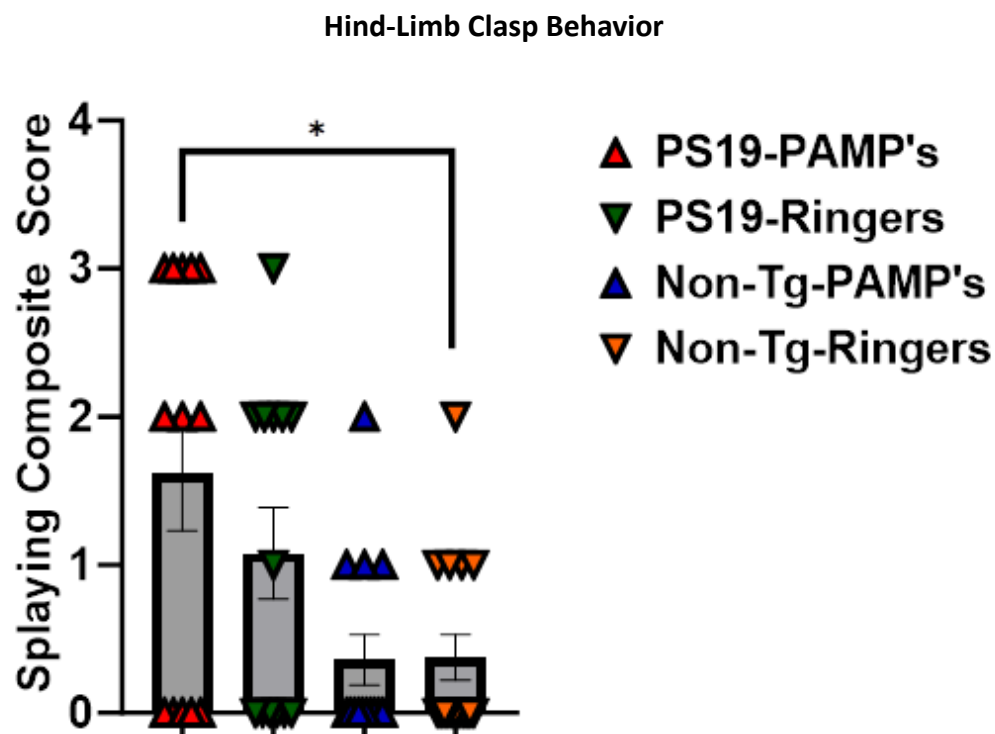


Figure 5: This graph depicts the hind-limb clasp behavioral test of PS19+ mice who received PAMP injections (**red** triangles), PS19+ mice who received mammalian ringers' solution (**green**

triangles), Non-Tg mice that received PAMPs (**blue** triangles), and Non-Tg mice that received Ringers (**orange** triangles). A parametric test was chosen for the analysis of the hind-limb clasp behavioral data because a Shapiro-Wilkes test of normality showed that the data was not normally distributed in the PS19+ PAMPs group ($W=.7539$, $P=.002$), or the PS19+ Ringers group ($W=.7858$, $P=.0046$), the Non-Tg PAMPs group ($W=.6269$, $P<.0001$) or the Non-Tg Ringers group ($W=.6483$, $P<.0001$).

An unpaired parametric T-test was used to determine behavioral differences in mice based on treatment effects and genotype using the open-field behavioral test. There was a significant difference in basic motor function (distance traveled) between treatment groups of PS19+ mice ($p=.0367$). PS19+ mice that received PAMPs had a median basic movement distance of 1,322 cms over a 5-minute trial ($n=15$). PS19+ mice that received Ringers had a median distance of 1,559 cms ($n=15$).

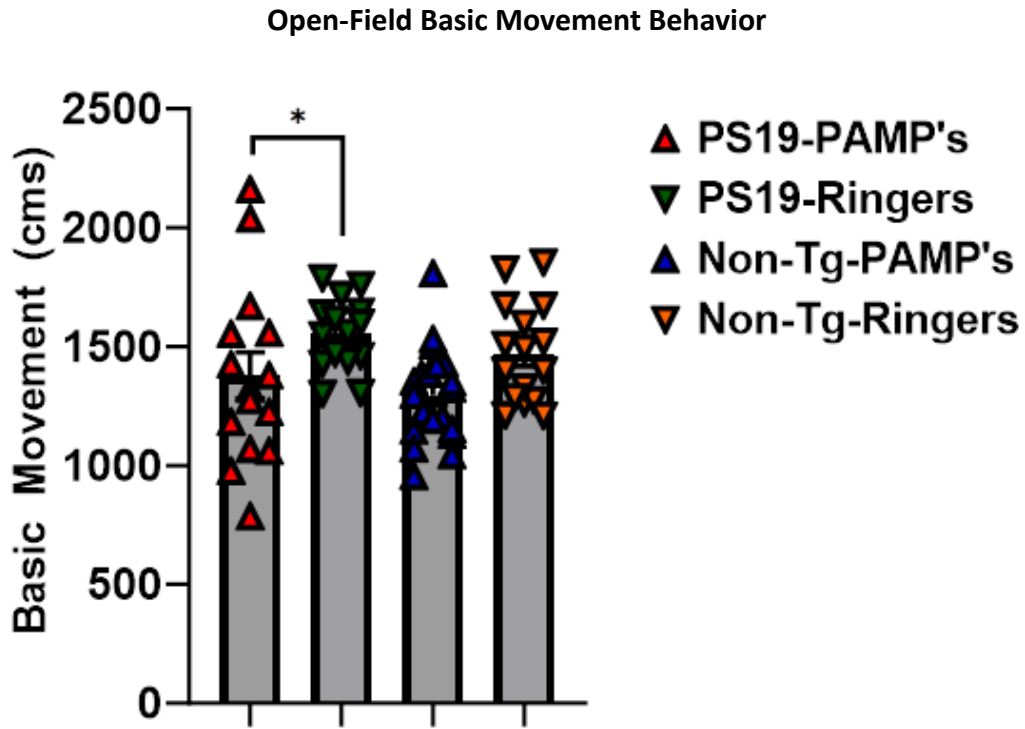


Figure 6: Graph depicting the open-field behavioral test basic movement (cms) of PS19+ mice who received PAMP injections (**red** triangles), and PS19+ mice who received Mammalian Ringers solution (**green** triangles), Non-Tg mice who received PAMPs (**blue** triangles), and Non-Tg mice who received Ringers (**orange** triangles).

An unpaired parametric T-test was used to determine that PS19+ mice that received PAMPs showed significant difference in rearing behavior compared to PS19+ mice that received only Ringers ($p = .0284$). The PS19+ PAMPs group had a mean of 141.5 seconds spent on their hind-legs, out of 5 minutes total per behavioral test ($n=15$), while PS19+ Ringer's mice had a mean of 167.7 seconds per trial ($n=15$).

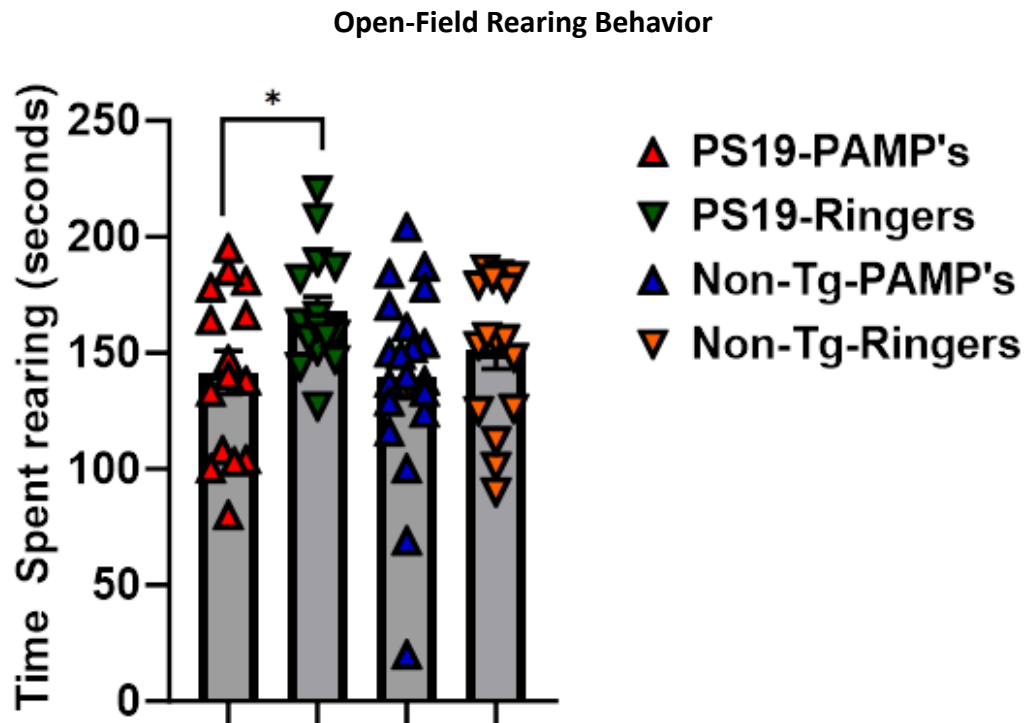


Figure 7: Graph depicting the open-field behavioral test rearing (seconds) of PS19+ mice who received PAMP injections (**red** triangles), and PS19+ mice who received Mammalian Ringers solution (**green** triangles), Non-Tg mice who received PAMPs (**blue** triangles), and Non-Tg mice who received Ringers (**orange** triangles).

A Pearson correlation matrix was used to determine the inter-test correlation of behavioral testing results for each mouse. Notably, Open-Field behavioral test basic movements were significantly correlated with rearing behavioral scores ($P=.004$), showing a positive and strong correlation between these two tests. Also significantly correlated, where rearing behavior and open-field behavioral testing fine movements ($P=.004$), showing a strong positive correlation.

Behavioral Testing Correlation Matrix

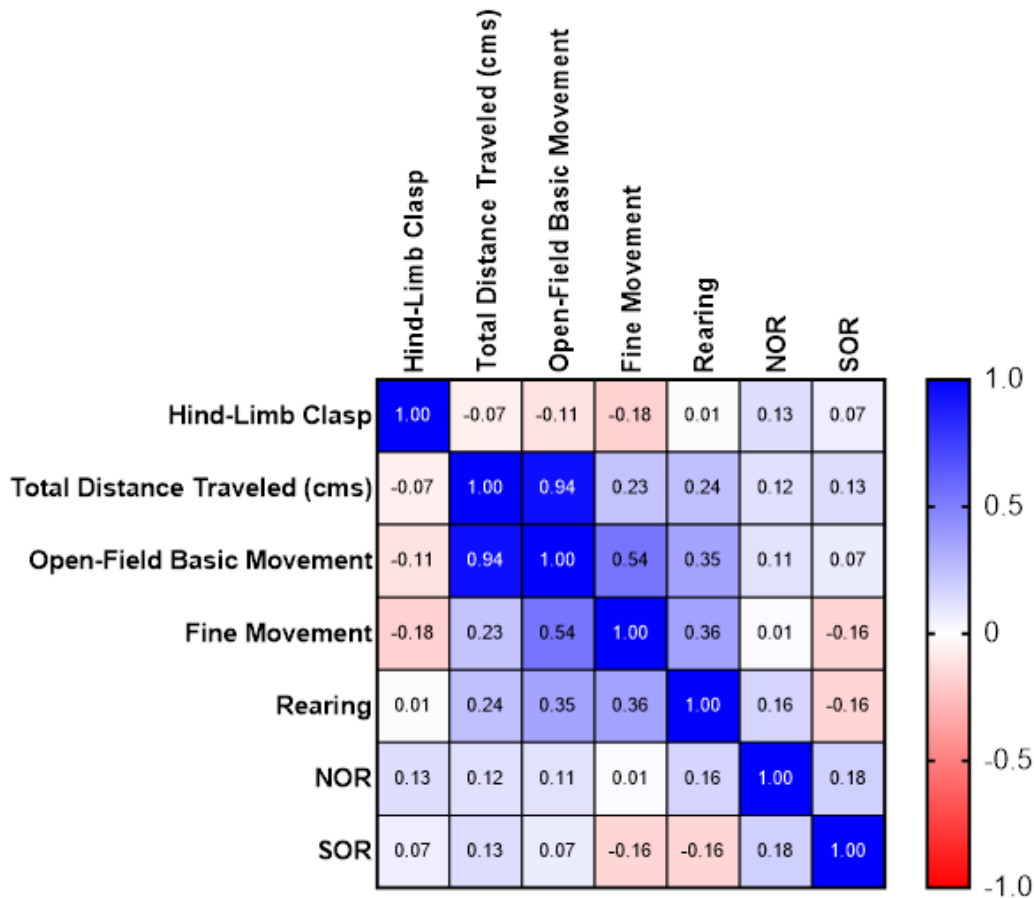


Figure 8: Pearson correlation matrix depicting the correlation between behavioral tests for each mouse. There was a low to moderate correlation between measures of open-field behavioral testing (total distance, basic movement, fine movement, rearing). There was a low positive correlation between NOR and SOR behavior results compared to other behavioral tests.

Histological Measures of Phosphorylated Tau and Total Humanized Tau

Positive staining was observed throughout the entire hippocampus, with the highest density in the CA1, subiculum, and dentate gyrus hilus areas (Fig 8A). Staining was also dense in the lateral and medial entorhinal cortical areas. An unpaired nonparametric (Mann-Whitney U) T-test was used to analyze the percentage of area covered by pathological tau, using the AT8

antibody for IHC. Quantification was performed using only the hippocampus region (Fig 8).

There was a significant difference in percentage of area covered between treatment groups;

PS19+ mice that received PAMPs showed significantly greater AT8 ($p = .049$) with a mean of

.6613% ($n=22$) compared to the control group that received mammalian Ringers with a mean of

0.2767% ($n=17$).

AT8 IHC Hippocampus

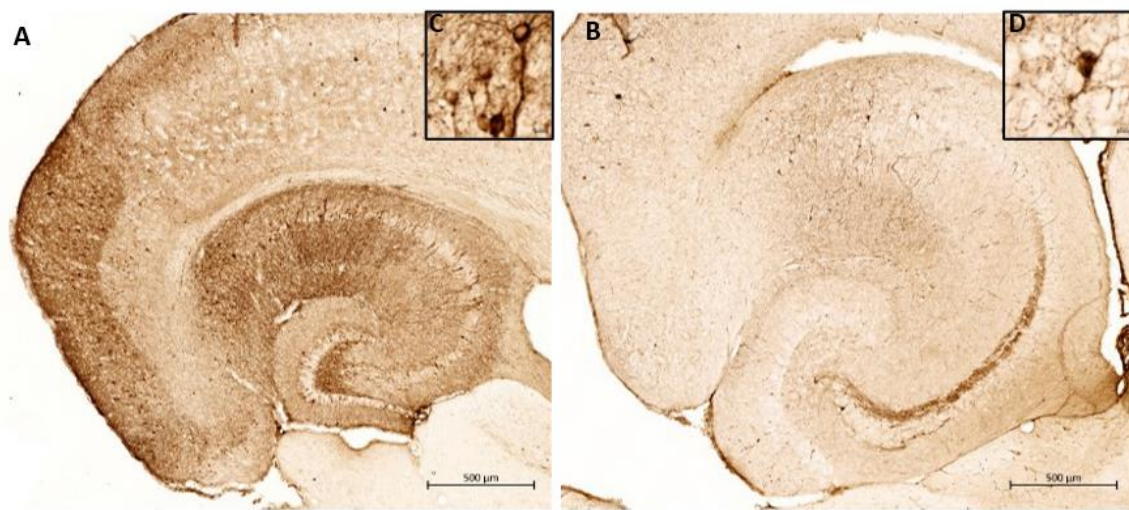


Figure 9: Pathological hyper-phosphorylated Tau (AT8) hippocampus IHC **A** PS19+ PAMP. , **C** is 20x magnification. **B** is PS19+ Mammalian Ringers (Control). Positive staining was sporadic in the CA1 region of the hippocampus, and the lateral entorhinal and perirhinal cortical areas. **D** is 20x magnification.

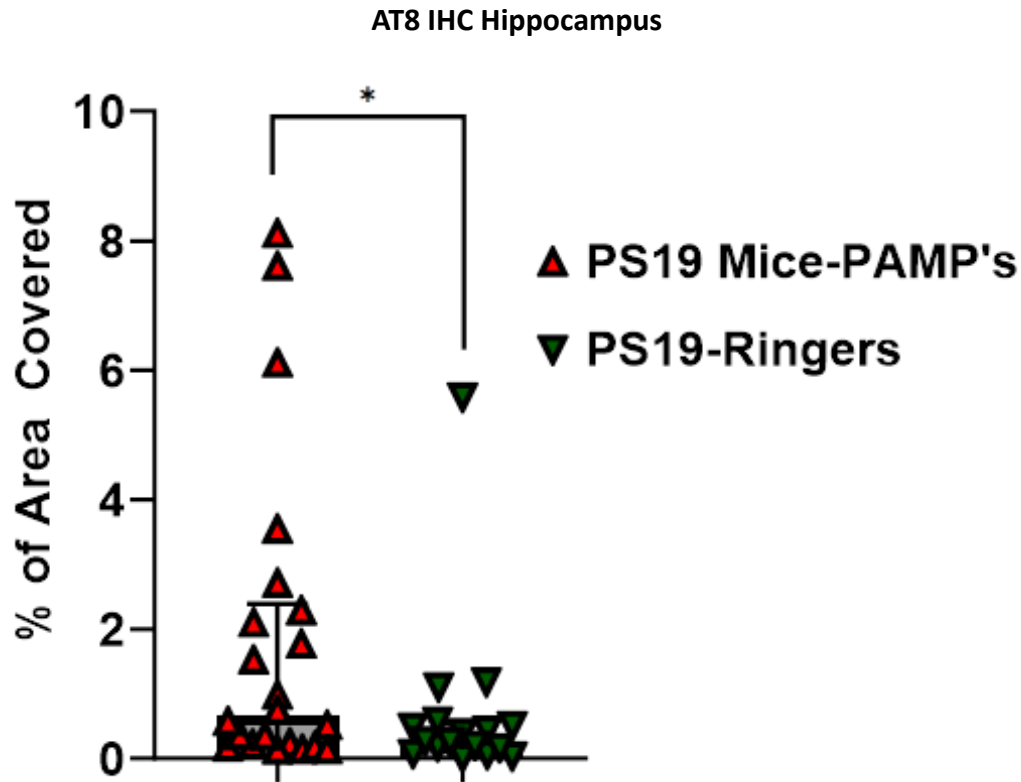


Figure 10: Graph depicting the % of area covered of the pathological tau (AT8) in the hippocampus of PS19+ mice who received PAMP injections (**red** triangles), and PS19+ mice who received mammalian Ringer's solution (**green** triangles). Non-parametric T-tests were used for statistical analysis because Shapiro-Wilk tests showed that IHC hippocampal AT8 measures were not normally distributed in the PS19+ PAMPs group ($W=.72$, $p < .0001$) or the PS19+ Ringers group ($W=.4765$, $p < .0001$). This graph depicts median values and Interquartile ranges (IQRs).

An unpaired nonparametric (Mann-Whitney U) T-test was also used to analyze the percentage of area covered by pathological tau using IHC methods, based on the AT8 antibody for the whole brain sections (Fig 9). There was a significant difference in percentage of area covered between treatment groups; PS19+ mice that received PAMPs showed significantly

greater % of area covered of AT8 ($p = .0063$) with a median of .2813% ($n=21$) compared to the control group that received mammalian Ringers with a median of .09375% ($n=17$) (Fig 10).

AT8 IHC Whole-Brain

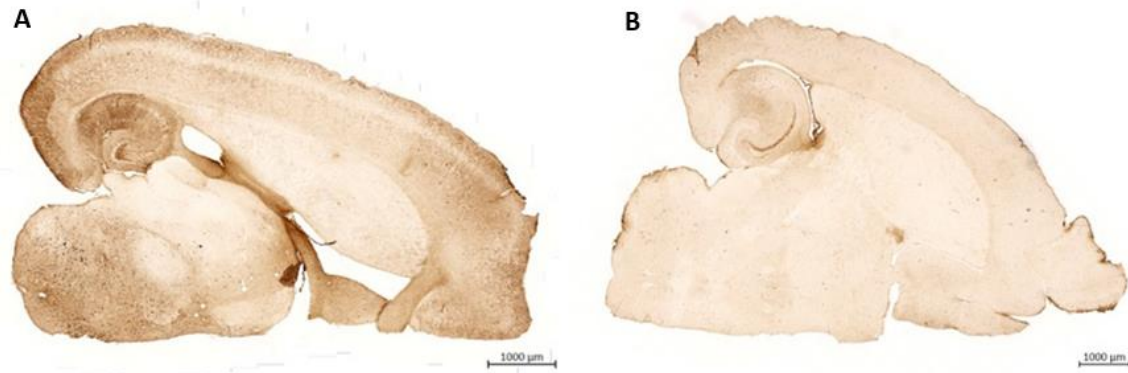


Figure 11: Pathological Tau (AT8) whole-brain IHC **A** PS19+ PAMP. Positive staining was observed in the hippocampus, brain stem, frontal cortex, perirhinal cortex, and the lateral and medial entorhinal cortex. **B** PS19+ Mammalian Ringers (Control). Sporadic staining was observed in the CA1 region of the hippocampus and entorhinal and perirhinal cortical areas.

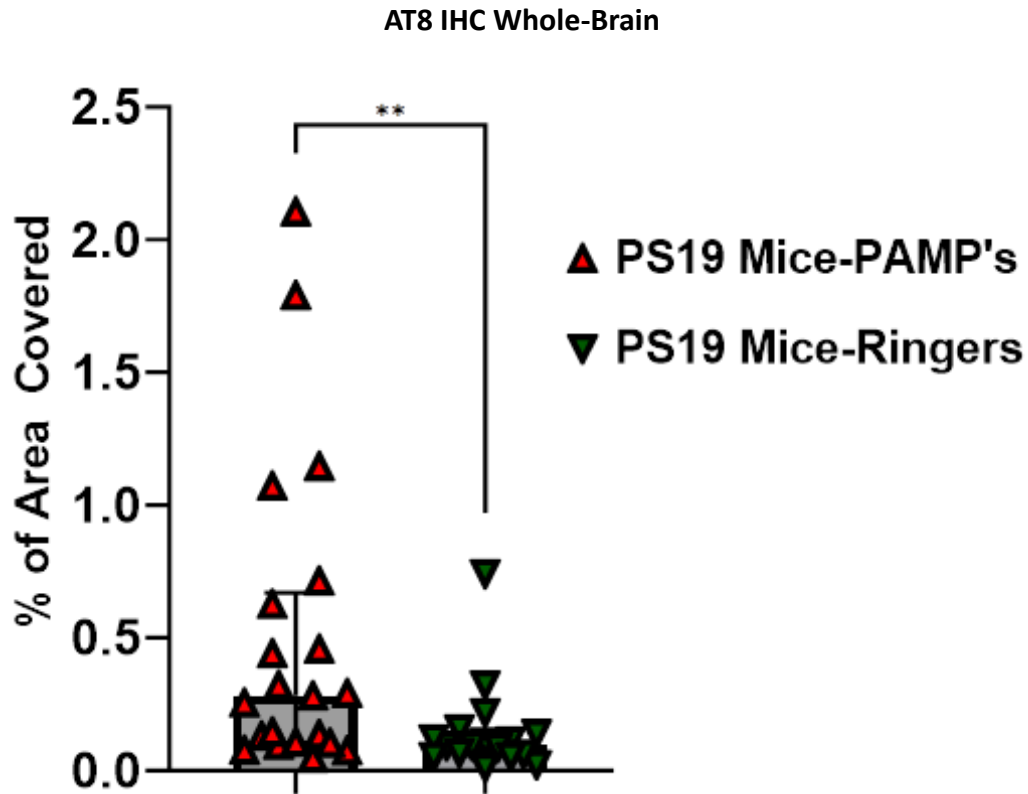


Figure 12: Graph depicting the % of area covered of the pathological tau (AT8) in the whole-brain of PS19+ mice who received PAMP injections (**red** triangles), and PS19+ mice who received Mammalian Ringers solution (**green** triangles). Non-parametric T-tests were used for statistical analysis because Shapiro-Wilk tests showed that IHC whole-brain AT8 measures were significantly non-normally distributed in the PS19+ PAMPs group ($W=.7488$, $p=.0001$) and the PS19+ Ringers group ($W=.6321$, $p < .0001$). The bar graphs represent median values per group and the error bars represent IQR for that group.

HT7 staining is most dense in the sub-granular layer of the dentate gyrus in the hippocampus (Fig 11). An unpaired nonparametric (Mann-Whitney U) T-test was used to analyze the percentage of area covered by total tau, using the HT7 antibody for IHC. Quantification was performed using only the hippocampal region of the brain (Fig 12). There

was no statistically significant difference in percentage of area covered between treatment groups ($p = .639$); PS19+ mice that received PAMPs had a median of 1.596% ($n=20$) compared to the control group that received Mammalian Ringers with a median of 1.048% ($n=18$).

HT7 IHC Hippocampus

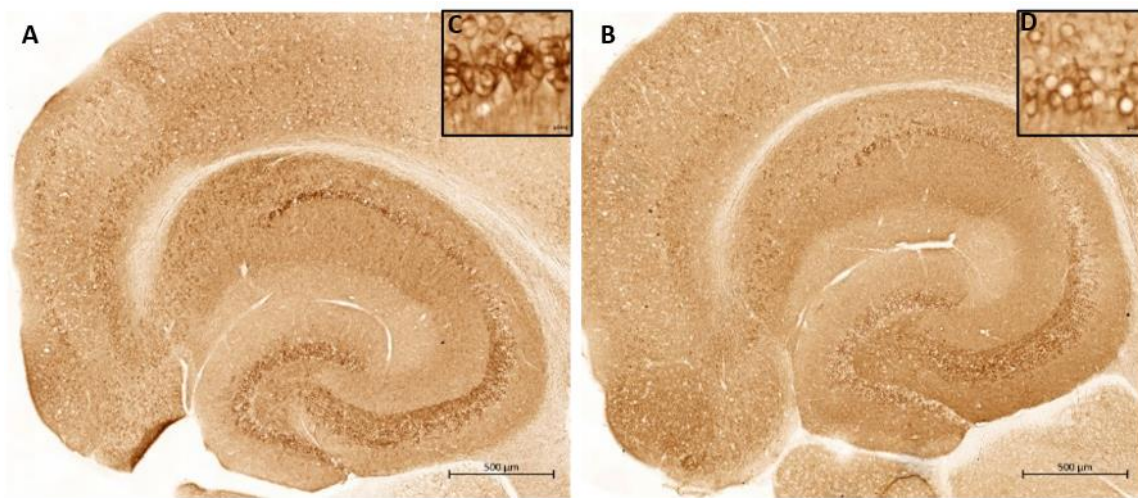


Figure 13: Total humanized tau (HT7) hippocampus IHC **A** PS19+ PAMP. **C** is 20x magnification. **B** is PS19+ Mammalian Ringers (Control). Positive staining is evenly distributed throughout the hippocampus. **D** is 20x magnification. Non-Tg mice were not included in IHC measures of total humanized tau because the HT7 antibody only measures the humanized form of tau, and Non-Tg mice did not contain any humanized tau (compared to PS19+ mice with the humanized transgene).

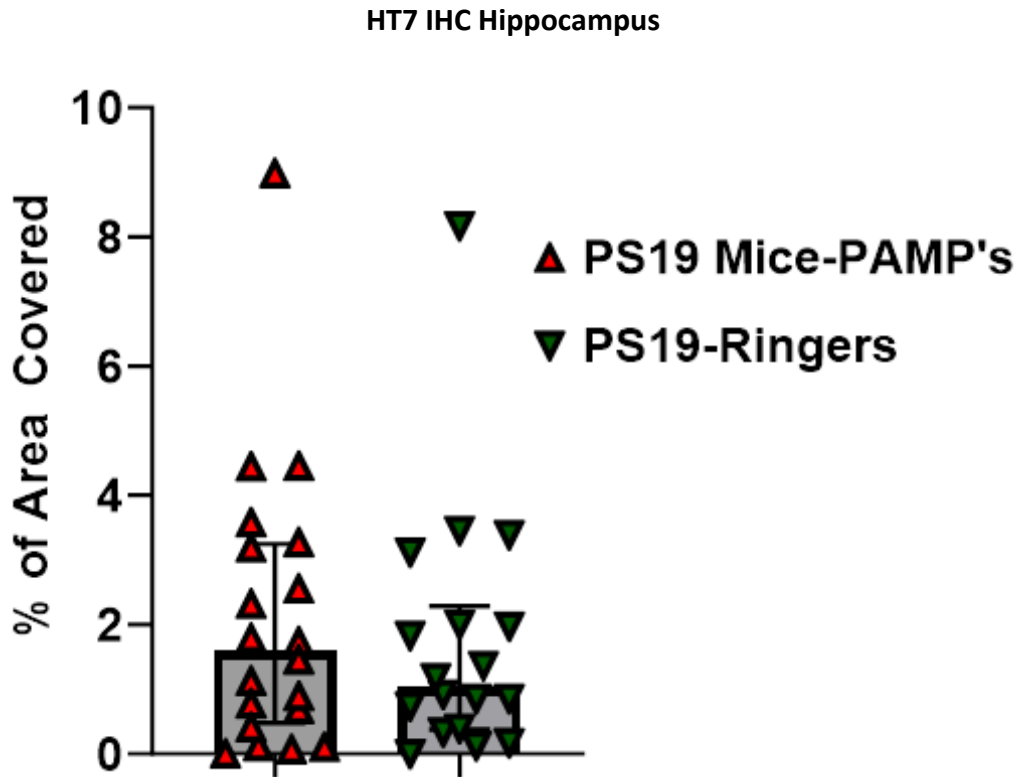


Figure 14: Graph depicting the % of area covered of the total humanized tau (HT7) in the hippocampus of PS19+ mice who received PAMP injections (**red** triangles), and PS19+ mice who received Mammalian Ringers solution (**green** triangles). Non-parametric T-tests were used for statistical analysis because Shapiro-Wilk tests showed that IHC hippocampal HT7 measures were significantly non-normally distributed in the PS19+ PAMPs group ($W=.831$, $p=.0026$) and the PS19+ Ringers group ($W=.7529$, $p=.0013$). The bar graphs represent median values per group and the error bars represent IQR for that group.

An unpaired nonparametric (Mann-Whitney U) T-test was used to analyze the percentage of area covered by total tau, using the HT7 antibody for IHC (Fig 13). This analysis was performed using whole-brain tissue sections. There was no significant difference in percentage of area covered between treatment groups ($p= .5728$); PS19+ mice that received

PAMPs had a median of .4114% (n=21) compared to the control group that received Mammalian Ringers with a median of .327% (n=18) (Fig 14).

HT7 IHC Whole-Brain

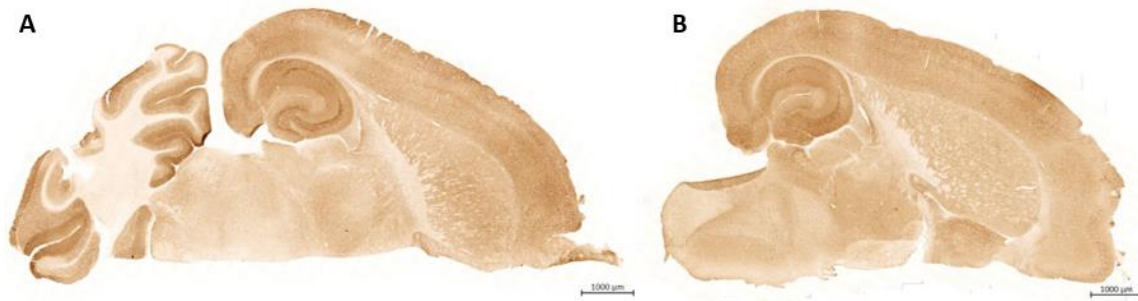


Figure 15: Total humanized tau (HT7) whole-brain IHC **A** PS19+ PAMP. Positive staining is most dense in the hippocampus, specifically the sub granular layer of the dentate gyrus. The cortical areas also display more intense staining compared to midbrain and brainstem areas. **B** PS19+ Mammalian Ringers (Control). The hippocampus shows dense staining, as well as the insular cortex.

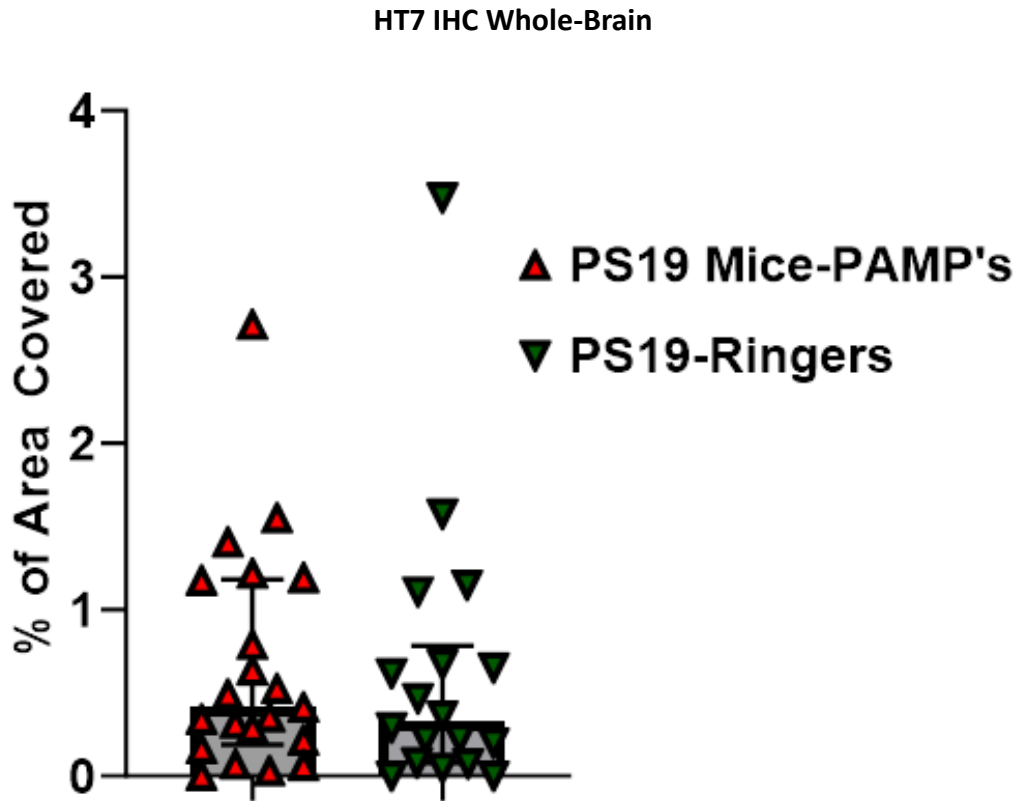


Figure 16: Graph depicting the % of area covered of the total humanized tau (HT7) in the whole-brain of PS19+ mice who received PAMP injections (**red** triangles), and PS19+ mice who received Mammalian Ringers solution (**green** triangles). Non-parametric T-tests were used for statistical analysis because Shapiro-Wilk tests showed that IHC whole-brain HT7 measures were not normally distributed in the PS19+ PAMPs group ($W=.8358$, $p=.0024$) and the PS19+ Ringers group ($W=.7006$, $p < .0001$). No significant differences were observed. The bar graphs represent median values per group and the error bars represent IQR for that group.

Histological Measures of Neuroinflammation

Dense GFAP staining is observed evenly throughout the hippocampus, with the exception of the sub-granular layer of the dentate gyrus in PS19 mice injected with PAMPs (Fig 15A). In PS19 mice injected with Ringers positive staining occurs throughout the hippocampus,

but less prominently in the pre-subiculum and peri-subiculum (Fig 15B). In Non-Tg PAMP treated mice positive staining is found throughout the hippocampus, but less severe in the pre-subiculum and the peri-subiculum (Fig 15 C). In Non-Tg mice treated with Ringer's, light staining is found throughout the hippocampus, with less staining in the pre-subiculum and peri-subiculum (Fig 15 D). Unpaired nonparametric (Mann-Whitney U) T-tests were performed to analyze the percentage of area covered by reactive astrocytes, using the GFAP antibody for IHC assays and subsequent analysis. This analysis was performed using only the hippocampal regions of the brain. There was not a statistically significant difference in percentage of area covered between treatment groups within the same genotype of PS19+ mice ($p = .1801$); PS19+ mice that received PAMPs showed a median of 10.63% ($n=22$) compared to PS19+ vehicle-only (Ringers) group, which had a median of 6.81% ($n=19$) (Fig 16). However, there was a significant effect of PAMP treatment in Non-Tg mice ($p = .009$). Non-Tg mice that received PAMPs had a median GFAP percentage of area covered of 8.38% ($n=22$), while the Non-Tg mice that received the vehicle (Ringers) had a median of 4.51% ($n=19$).

There was no significant difference in percentage of area positive for GFAP in the hippocampus between genotypes in the groups receiving PAMPs ($p = .1789$). PS19+ mice that received PAMPs had a median of 10.63% ($n=22$), while Non-Tg mice that received PAMPs had a median of 8.38% ($n=22$). Similarly, there was no significant difference between genotypes for mice receiving the vehicle (Ringers) ($p = .0906$). PS19+ mice that received Ringers had a median of 6.81% ($n=19$), while Non-Tg mice that received Ringers had a median of 4.51% ($n=19$). PS19+ mice that received Ringers showed no significant difference ($p = .0906$) compared to Non-Tg

mice that received Ringers. PS19+ mice that received Ringers had a median % of area covered of 6.81% (n=19), while Non-Tg mice that received Ringers showed a median of 4.51% (n=19).

GFAP IHC Hippocampus

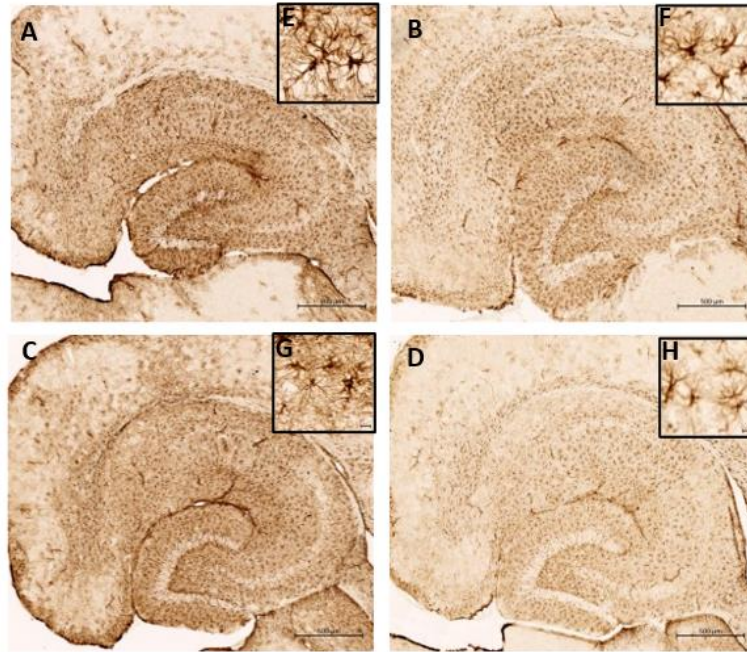


Figure 17: GFAP hippocampus IHC. **A** PS19+ PAMP. **E** is a 20x objective magnification. **B** PS19+ Ringers. **F** is a 20x objective magnification. **C** Non-Tg PAMP. **G** is a 20x objective magnification. **D** Non-Tg Ringers. **H** is a 20x objective magnification.

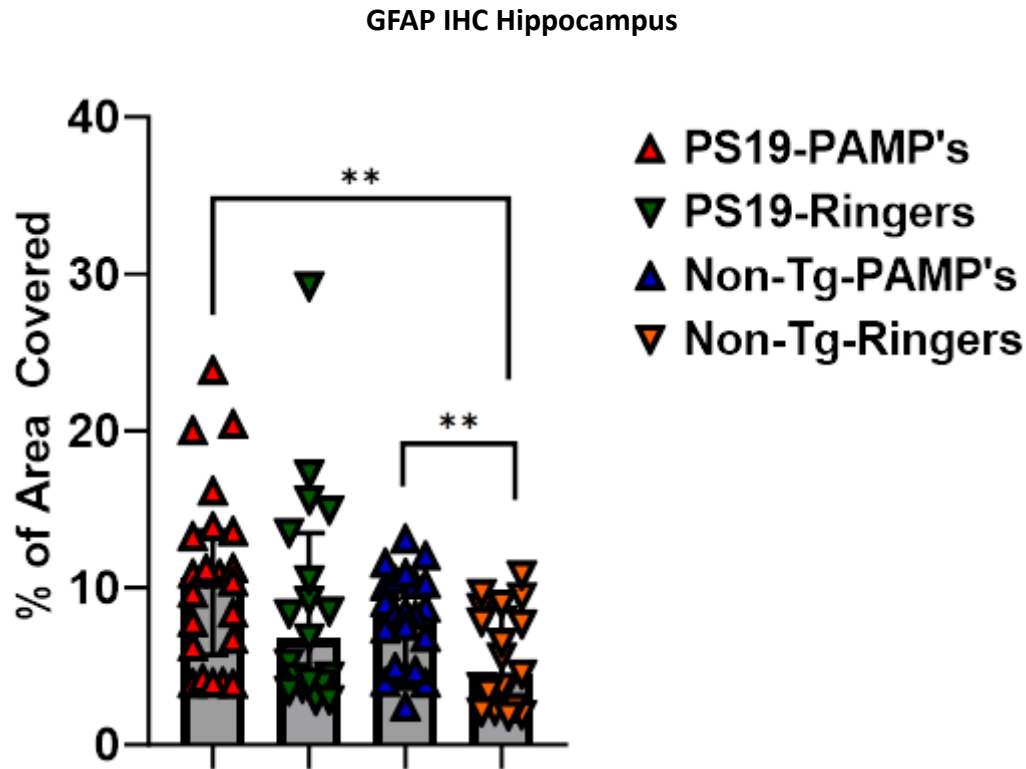


Figure 18: Graph depicting the % of area covered of reactive astrocytes (GFAP) in the hippocampus of PS19+ mice who received PAMP injections (**red** triangles), and PS19+ mice who received Mammalian Ringers solution (**green** triangles), Non-Tg mice who received PAMPs (**blue** triangles), and Non-Tg mice who received Ringers (**orange** triangles). Non-parametric T-tests were used for statistical analysis because Shapiro-Wilk tests showed that IHC hippocampal GFAP measures were significantly non-normally distributed in the PS19+ Ringers group ($W=.8157$, $p=.0019$), and the Non-Tg Ringers group ($W=.8941$, $p=.0381$). The bar graphs represent median values per group and the error bars represent IQR for that group.

In whole brain section analysis, an unpaired nonparametric (Mann-Whitney U) T-test was performed to analyze the percentage of area covered by reactive astrocytes, using the GFAP antibody for IHC assays and subsequent analysis (Fig 18). There was a statistically significant difference in percentage of area covered between treatment groups within the same genotype;

PS19+ mice that received PAMPs showed significantly greater GFAP ($p = .0226$) with a median of 5.22% ($n=22$) compared to the PS19+ control group that received Mammalian Ringers with a median of 3.35% ($n=19$). There was also a significant treatment effect between Non-Tg mice ($p = .0095$). Non-Tg mice that received PAMPs had a median GFAP percentage of area covered of 4.19% ($n=22$), while the Non-Tg mice that received the vehicle (Ringers) had a median of 1.84% ($n=19$).

There was no significant difference in percentage of area positive for GFAP between genotypes in the groups receiving PAMPs ($p = .0845$). PS19+ mice that received PAMPs had a median of 5.22% ($n=22$), while Non-Tg mice that received PAMPs had a median of 4.19% ($n=22$). Similarly, there was no significant difference between genotypes for mice receiving the vehicle (Ringers) ($p = .1025$). PS19+ mice that received the vehicle had a median of 3.35% ($n=19$), while Non-Tg mice that received the vehicle (Ringers) had a median of 1.84% ($n=19$).

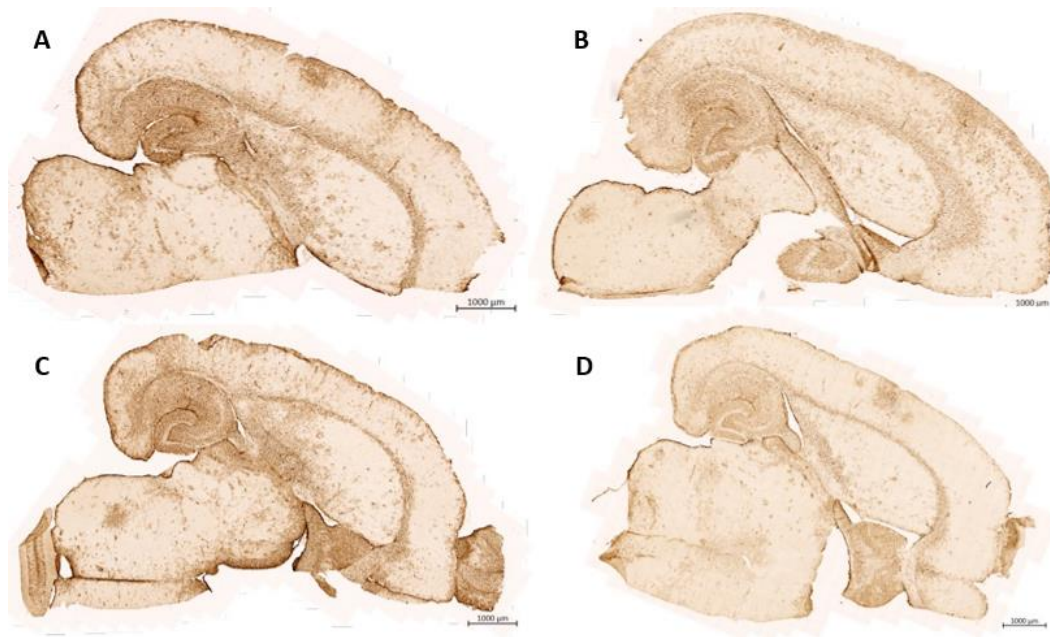
GFAP IHC Whole-Brain

Figure 19: Glial fibrillary acidic protein (GFAP) in whole-brain IHC sections. **A** PS19+ PAMP.

Staining is most dense in the hippocampus and the entorhinal cortex. **B** PS19+ Ringers. Positive staining is most dense in the hippocampal region and the ventro orbital cortex. **C** Non-Tg PAMP.

Staining is most dense in the hippocampal region. **D** Non-Tg Ringers. Light staining is found throughout the brain, with the highest density in the hippocampal region and insular area.

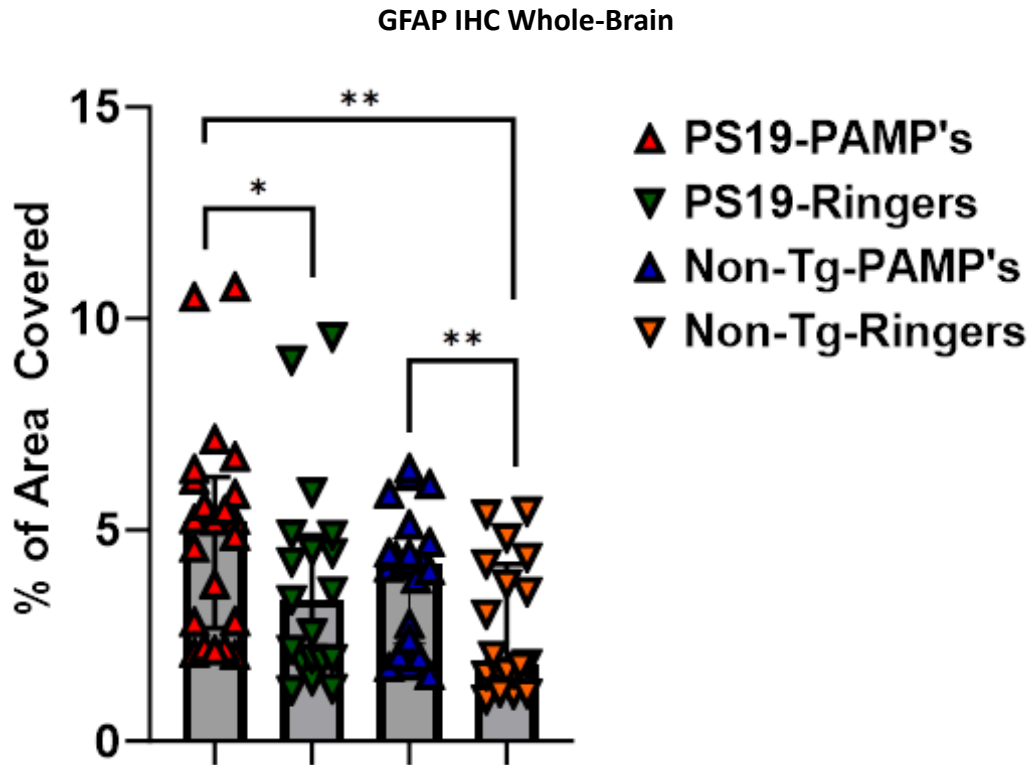


Figure 20: Graph depicting the % of area covered of reactive astrocytes (GFAP) in the whole-brain of PS19+ mice who received PAMP injections (**red** triangles), and PS19+ mice who received mammalian Ringer's solution (**green** triangles), Non-Tg mice who received PAMPs (**blue** triangles), and Non-Tg mice who received Ringers (**orange** triangles). Non-parametric T-tests were used for statistical analysis because Shapiro-Wilk tests showed that IHC hippocampal GFAP measures were significantly non-normally distributed in the PS19+ PAMPs group ($W=.8962$, $p=.025$), the PS19+ Ringers group ($W=.8539$, $p=.0078$), and Non-Tg Ringers group ($W=.8617$, $p=.0105$). The bar graphs represent median values per group and the error bars represent IQR for that group.

For MHC-II immunostaining in PAMP treated PS19 mice, the densest staining is found in the CA1 region of the hippocampus, as well as some sporadic staining the hilus of the dentate gyrus (Fig 19A). The entorhinal, perirhinal, and lateral entorhinal cortex also display staining. In

Ringer's treated PS19 mice sporadic staining is found in the hippocampus, primarily in the parasubiculum area (Fig 19 B). In Non-Tg mice staining is sparse, with a few positive cells in the hilus area of the dentate gyrus (Figs 19 C and D).

Unpaired nonparametric (Mann-Whitney U) T-tests were performed to analyze the percentage of area covered by MHC II for IHC assays and analysis (Fig 20). This analysis was performed using only the hippocampal region of the brain. There was a statistically significant difference in percentage of area covered between treatment groups within the same genotype of PS19+ mice ($p = .0015$); PS19+ mice that received PAMPs covered a median area of .1149% ($n=21$) compared to the PS19+ Ringers (control) group with a median of .05753% ($n=20$). There was also a statistically significant treatment effect between Non-Tg mice ($p = .007$); Non-Tg mice that received PAMPs had a median MHC II area of .06032% ($n=22$), while the Non-Tg mice that received the vehicle (Ringers) had a median area of .03461% ($n=19$).

There was a significant difference in percentage of area for MHC II between genotypes in the hippocampal region in mice that received PAMPs ($p < .0001$). PS19+ mice that received PAMPs had a median of .1149% ($n=21$), while Non-Tg mice that received PAMPs had a median of .06032% ($n=22$). There was also a significant difference of percentage of area covered by MHC II in the hippocampus between genotypes for mice receiving Ringers ($p = .0283$). PS19+ mice that received Ringers had a median of .05753% ($n=20$), while Non-Tg mice that received Ringers had a median of .03461% ($n=19$).

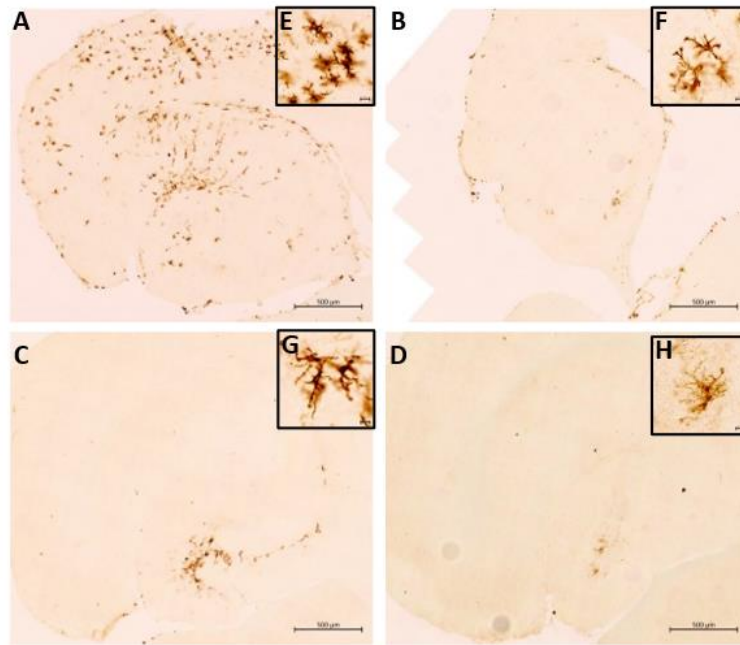
MHC II IHC Hippocampus

Figure 21: MHC II hippocampus IHC. **A** PS19+ PAMP. **E** is a 20x objective magnification. **B** PS19+ Ringers. Sporadic staining is found in the hippocampus, primarily in the parasubiculum area. **F** is a 20x objective magnification. **C** Non-Tg PAMP. **G** is a 20x objective magnification. **D** Non-Tg Ringers. **H** is a 20x objective magnification.

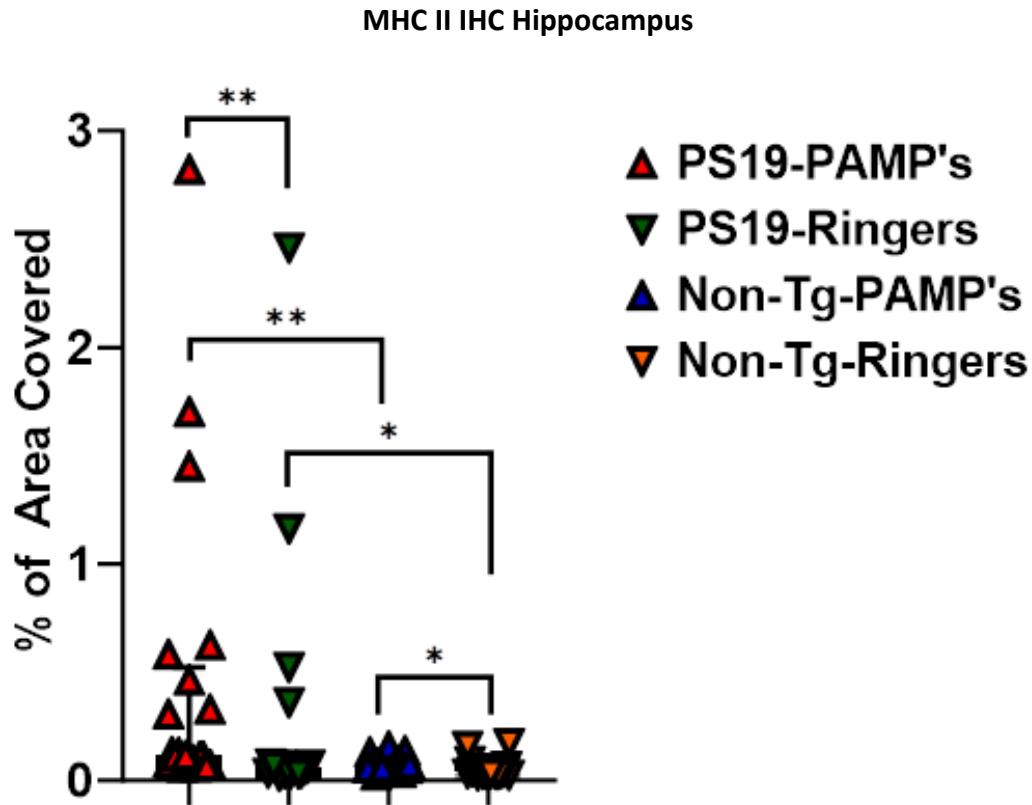


Figure 22: MHC II Hippocampal IHC results. PS19+ PAMP (red triangles), PS19+ Ringers (green triangles), Non-Tg PAMP (blue triangles), Non-Tg Ringers (orange triangles). Non-parametric T-tests were used for statistical analysis because Shapiro-Wilk tests showed that IHC hippocampal MHC II measures were significantly non-normally distributed in the PS19+ PAMPs group ($W=.6124$, $p < .0001$), the PS19+ Ringers group ($W=.4642$, $p < .0001$), the Non-Tg PAMPs group ($W=.8685$, $p=.0074$), and Non-Tg Ringers group ($W=.6496$, $p < .0001$). The bar graphs represent median values per group and the error bars represent IQR for that group.

Unpaired nonparametric (Mann-Whitney U) T-tests were performed to analyze the percentage of area covered by MHC II for IHC assays and analysis. This analysis was performed using whole-brain tissue sections. There was a statistically significant difference in percentage of area covered between treatment groups within the same genotype of PS19+ mice ($p = .0136$);

PS19+ mice that received PAMPs covered a median area of 0.07924% (n=19) compared to the PS19+ control group that received Mammalian Ringers with a median of 0.05276% (n=16). However, there was not a statistically significant treatment effect between Non-Tg mice based on treatment ($p = .7076$); Non-Tg mice that received PAMPs had a median MHC II area of 0.05697% (n=22), while the Non-Tg mice that received the vehicle (Ringers) had a median area of 0.0498% (n=19).

There was a significant difference in percentage of area positive for MHC II between genotypes in the groups receiving PAMPs ($p = .0041$). PS19+ mice that received PAMPs had a median of .07924% (n=19), while Non-Tg mice that received PAMPs had a median of .05697% (n=22). There was no significant difference of percentage of area covered by MHC II between genotypes for mice receiving Ringers ($p = .725$). PS19+ mice that received Ringers (control) had a median of .05276% (n=16), while Non-Tg mice that received the Ringers also had a median of .0498% (n=19).

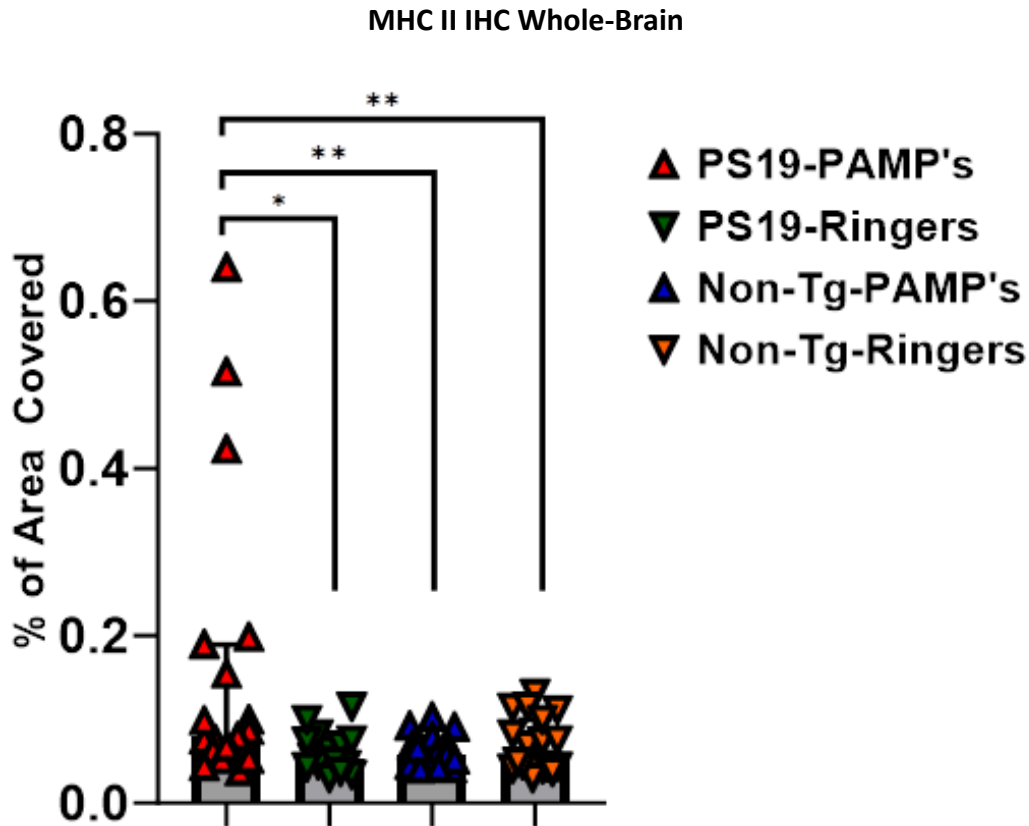


Figure 23: MHC II whole-brain IHC results. PS19+ PAMP (red triangles), PS19+ Ringers (green triangles), Non-Tg PAMP (blue triangles), Non-Tg Ringers (orange triangles). Non-parametric T-tests were used for statistical analysis because Shapiro-Wilk tests showed that IHC hippocampal MHC II measures were significantly non-normally distributed in the PS19+ PAMPs group ($W=.6746$, $p < .0001$), the Non-Tg PAMPs group ($W=.8826$, $p=.0135$), and Non-Tg Ringers group ($W=.868$, $p=.0133$). The bar graphs represent median values per group and the error bars represent IQR for that group.

For CD68 immunostaining, positive staining is present throughout the hippocampus, specifically in the CA1 and hilus of the DG. Intense staining is present in the entorhinal and perirhinal cortical areas (Fig 22). Unpaired nonparametric (Mann-Whitney U) T-tests were performed to analyze IHC results of the hippocampal-region only, based on the percentage of

area covered by CD68. There was no statistically significant difference in percentage of area covered between treatment groups within the same genotype of PS19+ mice ($p = .5088$); PS19+ mice that received PAMPs covered a median area of 1.419% ($n=17$) compared to the PS19+ Ringers (control) group with a median of .9288% ($n=13$). Similarly, there was not a statistically significant treatment effect between Non-Tg mice ($p = .4844$); Non-Tg mice that received PAMPs had a median CD68 area of 1.445% ($n=18$), while the Non-Tg mice that received the vehicle (Ringers) had a median area of 1.091% ($n=16$).

There was no significant difference in percentage of area for CD68 between genotypes in the groups receiving PAMPs ($p = .3865$). PS19+ mice that received PAMPs had a median of 1.419% ($n=17$), while Non-Tg mice that received PAMPs had a median of 1.445% ($n=18$). There was no significant difference of percentage of area covered of CD68 between genotypes for mice receiving Ringers ($p = .8799$). PS19+ mice that received Ringers (control) had a median of .9288% ($n=13$), while Non-Tg mice that received the Ringers had a median of 1.091% ($n=16$).

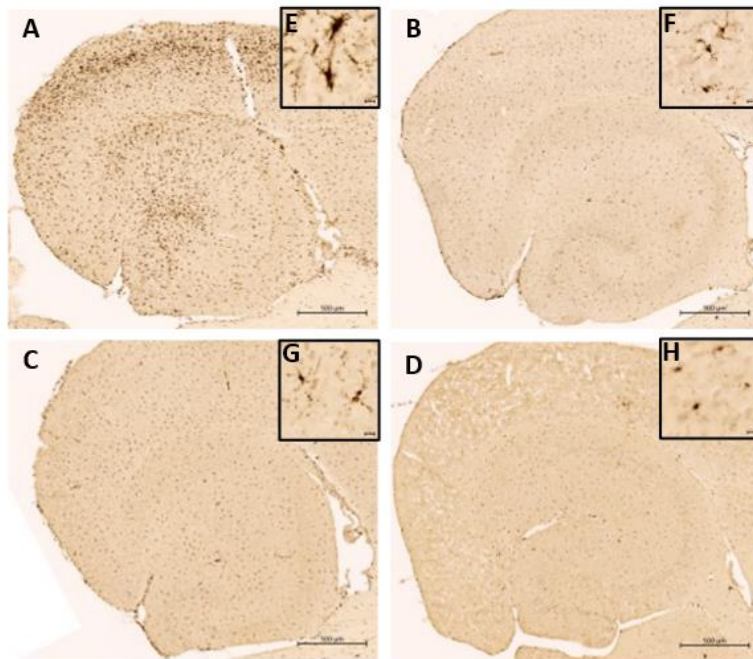
CD68 IHC Hippocampus

Figure 24: Graph depicting the % of area covered of CD68 in the hippocampus of **A.** PS19+ mice who received PAMP injections. **E.** with 20x objective magnification. **B.** PS19+ mice who received Mammalian Ringers solution. Light but even staining is present throughout the hippocampus **F.** with 20x objective magnification. **C.** Non-Tg mice who received PAMP injections. Staining is light but evenly distributed throughout the hippocampus. **G.** with 20x objective magnification. **D.** Non-Tg mice that received Mammalian Ringers solution. Staining is light throughout the hippocampus. **H.** with 20x objective magnification.

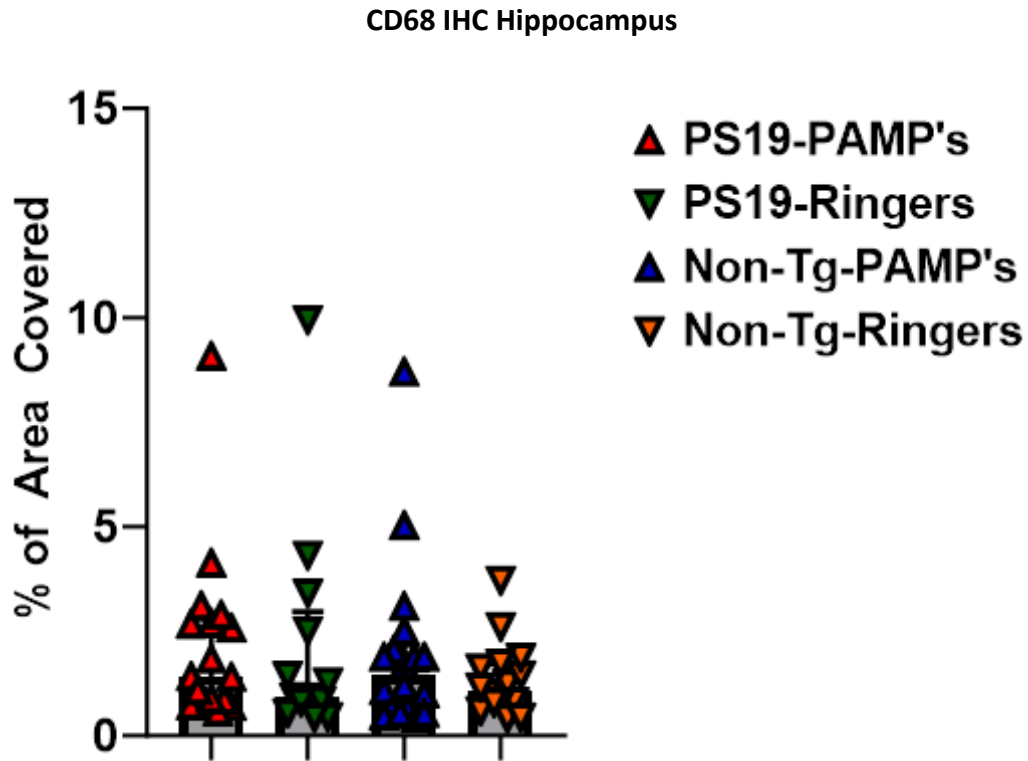


Figure 25: CD68 Hippocampal IHC results. PS19+ PAMP (red triangles), PS19+ Ringers (green triangles), Non-Tg PAMP (blue triangles), Non-Tg Ringers (orange triangles). Non-parametric T-tests were used for statistical analysis because Shapiro-Wilk tests showed that IHC hippocampal MHC II measures were significantly non-normally distributed in the PS19+ PAMPs group ($W=.7123$, $p=.0002$), the PS19+ Ringers group ($W=.6651$, $p=.0003$), the Non-Tg PAMPs group ($W=.7048$, $p < .0001$), and Non-Tg Ringers group ($W=.8546$, $p=.0159$). The bar graphs represent median values per group and the error bars represent IQR for that group.

Unpaired nonparametric (Mann-Whitney U) T-tests were performed to analyze whole-brain IHC results of the percentage of area covered by CD68. There was not a statistically significant difference in percentage of area covered between treatment groups within the same genotype of PS19+ mice ($p = .8691$); PS19+ mice that received PAMPs covered a median area of 1.296% ($n=17$) compared to the PS19+ Ringers (control) group with a median of .7356% ($n=13$).

There was a statistically significant treatment effect between Non-Tg mice ($p = .0358$); Non-Tg mice that received PAMPs had a median CD68 area of 1.272% ($n=18$), while the Non-Tg mice that received the Ringers had a median area of .8154% ($n=22$).

No significant difference was found for the percentage of area for CD68 between genotypes in the groups receiving PAMPs ($p = .8323$). PS19+ mice that received PAMPs had a median of 1.296% ($n=17$), while Non-Tg mice that received PAMPs had a median of 1.272% ($n=18$). There was no significant difference of percentage of area covered of CD68 between genotypes for mice receiving Ringers ($p = .331$). PS19+ mice that received Ringers (control) had a median of .7356% ($n=13$), while Non-Tg mice that received the Ringers had a median of .8154% ($n=22$).

CD68 IHC Whole-Brain

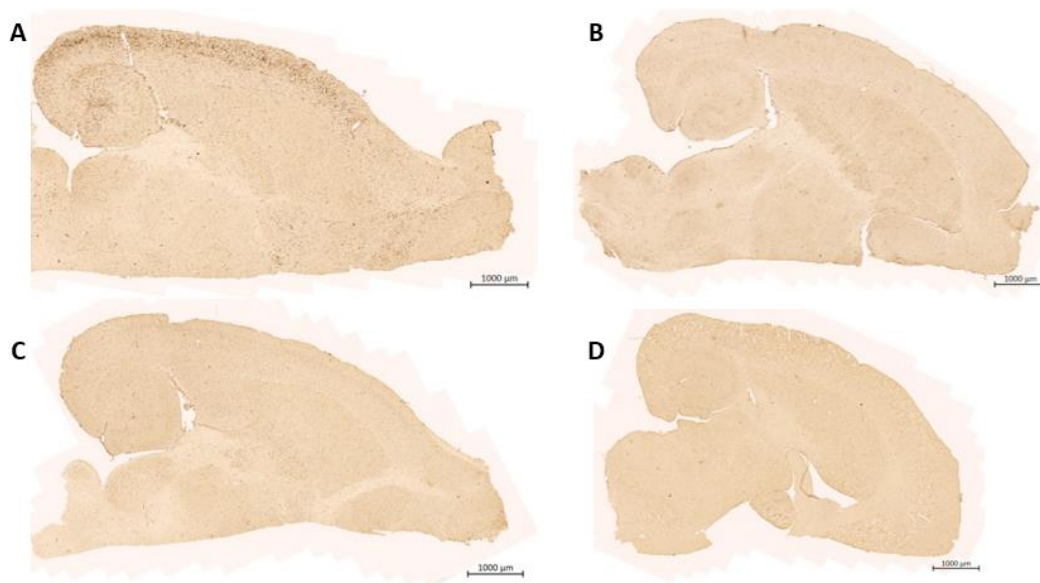


Figure 26: CD68 whole-brain IHC sections. **A** PS19+ PAMP. Positive staining is the most-dense in the hippocampus and entorhinal and entorhinal cortex. There is also dense staining in the insular cortex. **B** PS19+ Ringers. Staining is evenly distributed throughout the brain. **C** Non-Tg

PAMP. Staining is most dense in the hippocampus. **D** Non-Tg Ringers. Staining is light and evenly distributed.

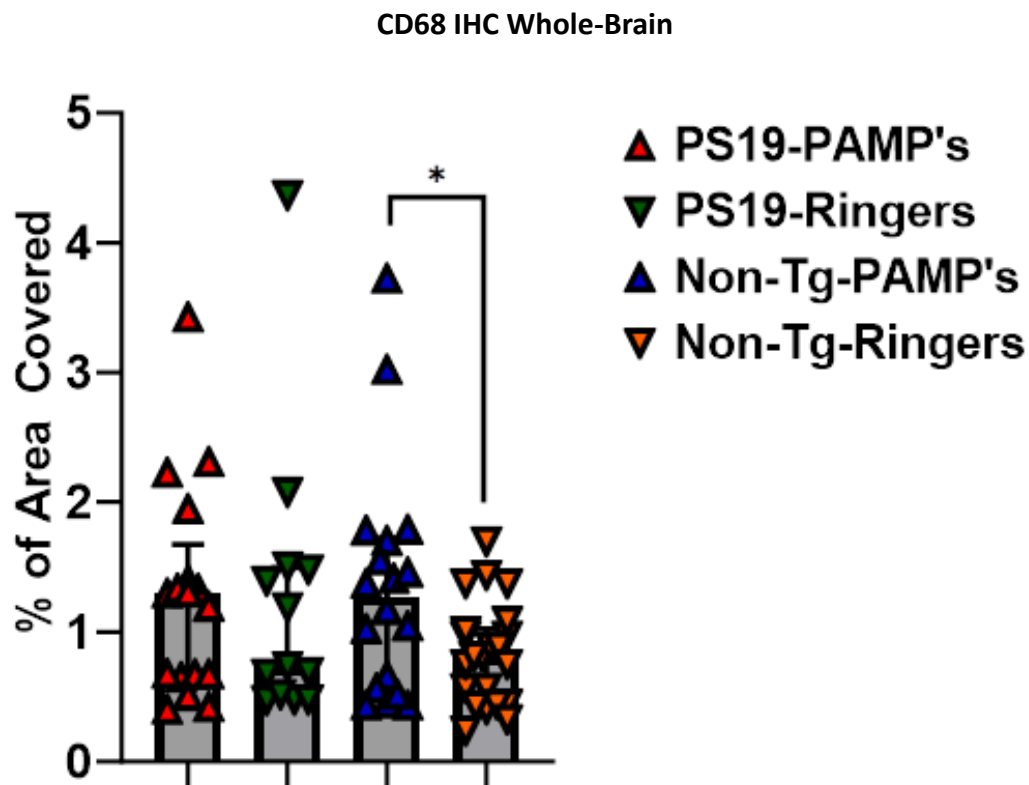


Figure 27: CD68 whole-brain IHC results. PS19+ PAMP (red triangles), PS19+ Ringers (green triangles), Non-Tg PAMP (blue triangles), Non-Tg Ringers (orange triangles). Non-parametric T-tests were used for statistical analysis because Shapiro-Wilk tests showed that IHC hippocampal MHC II measures were significantly non-normally distributed in the PS19+ PAMPs group ($W=.8746$, $p=.026$), the PS19+ Ringers group ($W=.7137$, $p=.0008$), and the Non-Tg PAMPs group ($W=.8533$, $p=.0095$). The bar graphs represent median values per group and the error bars represent IQR for that group.

A Spearman correlational matrix analysis was performed to determine potential differences between different markers of tau and inflammation for each PS19+ mouse in

hippocampal tissue section IHC results. There was a significant positive correlation between AT8 IHC results and MHC II results ($r=.502$, $P=.001$, $n=38$). There was also a positive significant correlation between AT8 % of area covered and CD68 % of area covered in the hippocampus ($r=.547$, $P=.002$, $n=29$). MHC II IHC % of area covered also showed a positive and significant correlation with CD68 hippocampal IHC values ($r=.402$, $P=.028$, $n=30$).

IHC % of Area Covered Hippocampus Correlation Matrix

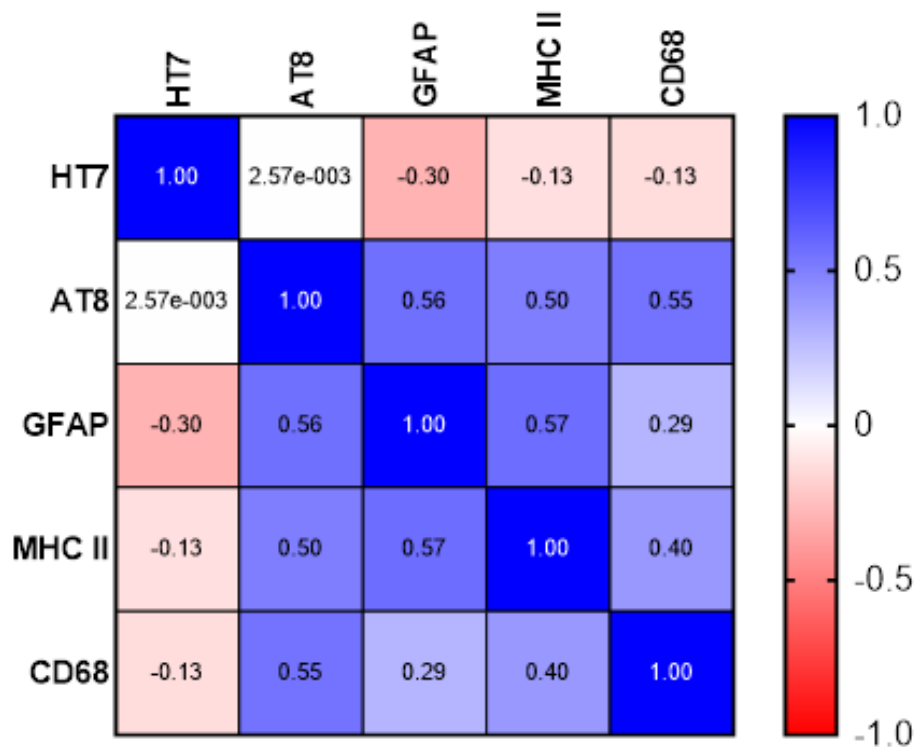


Figure 28: Spearman correlation matrix for IHC markers using PS19+ mice in hippocampal tissue.

There was a moderate to high positive correlation between pathological tau (AT8) and immune cell markers (GFAP, MHC II, CD68). There was a moderate to high positive correlation between immune cell markers. There was a low negative correlation between total humanized tau (HT7) and immune cell markers. The Spearman correlation matrix test was used because of the previously described non-normal distribution of IHC data.

A Spearman correlational matrix analysis was performed to determine potential differences between different markers of tau and inflammation for each mouse in whole-brain IHC results. AT8 was positively and significantly correlated with GFAP IHC result ($r=.503$, $P=.001$, $n=38$). AT8 was also positively correlated with MHC II IHC results, showing a positive and significant correlation ($r=.486$, $P=.002$, $n=37$).

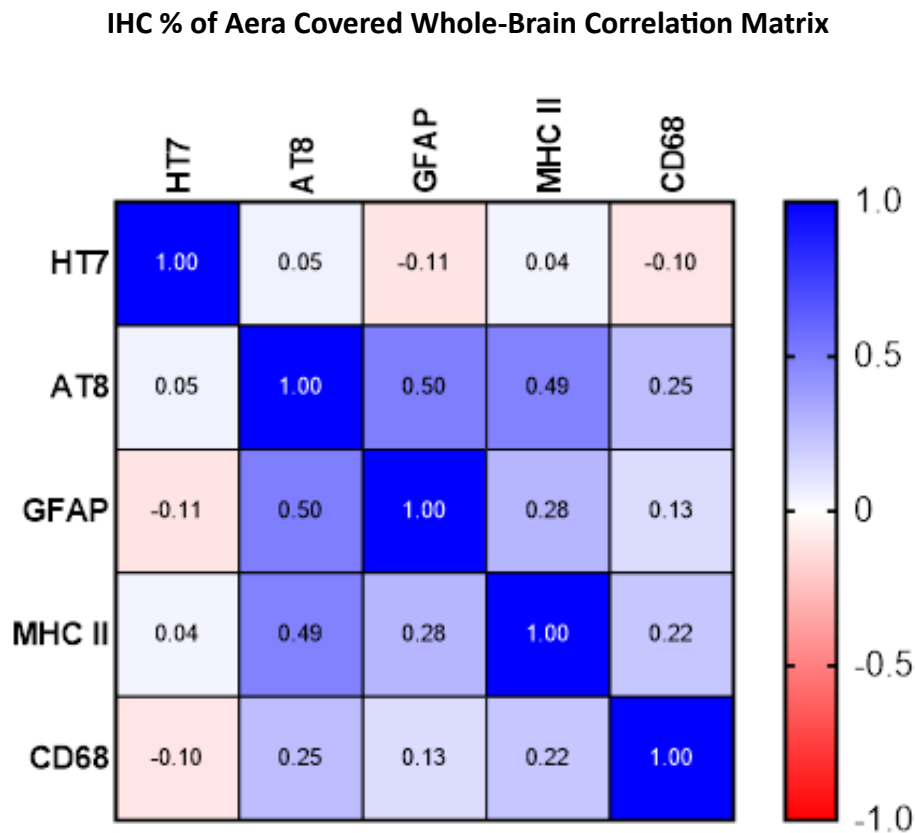


Figure 29: Spearman correlation matrix for IHC markers using PS19+ mice in whole-brain analysis. There was a moderate to strong positive correlation between pathological tau (AT8) and immune system markers (GFAP, MHC II, CD68). There was a moderate to strong positive correlation between immune cell markers (GFAP, MHC II, CD68). Total humanized tau (HT7) did not correlate with pathological tau or immune cell markers. The Spearman correlation matrix test was used because of the previously described non-normal distribution of IHC data.

Discussion

PAMPs Increase Pathological Tau Burden in PS19 Mice

Mice that received PAMPs showed significantly greater pathological tau burden in IHC results (AT8) compared to mice that received only the vehicle (Ringers). This effect was found in both whole-brain and hippocampal-only tissue. These results lend further support to evidence that inflammatory stress contributes to hyperphosphorylation of MAPT, which subsequently affects tau deposition—eventually leading to synaptic dysfunction and neuronal loss in AD (Sanabria-Castro et al., 2017).

IHC quantification for total tau (HT7) was not significantly different between the PAMP and control (Ringers) groups. These results suggest that total tau quantity is not significantly altered by PAMP-induced inflammatory stress, but that only pathological tau levels are primarily affected downstream through hyperphosphorylation. Hyperphosphorylation of tau results in the dissociation of MAPT from microtubules into soluble tau, which is not associated with NFTs (Reiss, 2021). Evidence suggests that soluble tau necessarily proceeds the conformational changes that take place for insoluble tau formation to occur (Hirata-Fukae et al., 2009). However, Hirata-Fukae and colleagues demonstrate that the mechanisms underlying the progression of soluble tau into insoluble tau, including paired helical filaments and subsequently NFTs, are still poorly understood. One explanation for significantly elevated pathological tau (AT8) with simultaneous non-significant (but still elevated) total tau (HT7) is the prion-like nature of pathological hyperphosphorylated tau. Pathological tau disassociates normal tau from the microtubule, creating a positive-feedback loop that ultimately produces more pathological tau from normal tau (Iqbal, 2005). Outside of the normal baseline production

and destruction of tau within each neuron, tau is not being created nor destroyed, but altered through dissociation from the microtubule and then conformationally into PHFs and NFTs.

Overall, these data suggest that PAMPs have a relatively selective influence on the insoluble pathological forms of tau compared to normal and soluble tau that makes up a functional microtubule. These pathological forms of tau create unstable microtubules that attenuate synaptic function and eventually lead to neurotoxicity.

PAMPs Contribute to a Pro-Inflammatory Environment

PS19+ mice that received PAMPs displayed significantly increased IHC markers of neuroinflammation compared to PS19+ mice that received only the vehicle (Ringers). PAMP treatment also significantly increased markers of the immune response associated with a pro-inflammatory state using IHC analysis.

Reactive astrocyte % of area covered based on IHC GFAP staining was significantly elevated throughout the whole brain in PS19+ mice injected with PAMPs, compared to PS19+ mice injected with only the control (Ringers). This was also found in the Non-Tg mice, suggesting that PAMPs affect astrocyte phenotype (reactivity) rather than mediation through increased pathological tau. Even though the sickness behavior caused by the PAMP injections resolved within a few days (data not shown), the elevations in markers of glial activation remained for at least a month. This implies the possibility that PAMPs can result in long lasting changes in the inflammatory tone of the brain that extend beyond the duration of the sickness response.

Elevated MHC II, a marker of inflammation that are presented on antigen presenting cells such as microglia in the brain, were significantly elevated based on treatment and genotype in both whole-brain and hippocampal IHC tissue analysis. MHC II has been shown to

be elevated in humans with AD—while they are constitutively expressed on antigen presenting cells, they can become upregulated in the presence of pathogens (Tooyama, 1990; Hopperton, 2018). MHC II is correlated with the disease-associated microglia (active) type of microglia, which consists of and produces a pro-inflammatory state through overproduction of pro-inflammatory cytokines (IL-1 β , IL-6, IL-18) and increased capacity for phagocytosis (Hansen, 2018). Notably, PS19+ mice that received PAMPs showed significantly elevated MHC II % of area covered compared to the PS19+ mice that received only the control (Ringers). This suggests that PAMPs play a role in altering microglia preferentially towards a pro-inflammatory state. This makes sense because PAMPs have been shown to induce inflammatory stress and alter microglia morphology (Saijo, 2013).

IHC analysis of CD68 did not show a statistically significant difference based on treatment or genotype. CD68 is a microglia marker that labels a receptor specific to macrophages (microglia) and dendritic cells, with most of the receptor being located intracellularly specifically on endosomes and lysosomes (Holness, 1992; Damjanac, 2007). Therefore, CD68 expression correlates with upregulation of phagocytic activity. Therefore, the lack of statistically significant differences in PS19+ mice based on treatment demonstrate that PAMPs may preferentially increase antigen presentation (MHC II) and cytokine production leading to a pro-inflammatory state, over phagocytic-type microglia. Non-Tg mice treated with PAMPs did display significantly greater % of area covered CD68 in whole-brain IHC analysis compared to Non-Tg mice that received only the control (Ringers). This suggests that the presence of pathological tau dampens the expression of CD68 in the presence of PAMPs. Alternatively, mice without the presence of pathological tau that received PAMPs did show a

significant increase in CD68 expression—suggesting greater phagocytic activity in relatively normal mice exposed to infection mimetics.

These results suggest that PAMPs play an active role in propagating the pro-inflammatory environment associated with AD. However, their role is complex, specifically related to microglia phenotype and activity. Microglia morphology polarization depends on many factors, including the type and presence of PAMPs, as well as the disease stage and the micro-environment and macro-environment in the brain (Walker, 2015).

Effects of PAMPs on Mouse Behavior

Given the effects of PAMPs in increasing pathological forms of tau, it is important to determine if this has functional consequences on the mice such as changes in behavior. PAMP-treated PS19 mice showed a significantly reduced amount of locomotor activity (in cms) in the open-field behavioral test compared to PS19+ Ringers injected mice. There was also a reduced amount of rearing in the PAMP treated PS19 mice compared to Ringers-injected animals. However, Non-Tg mice displayed no treatment-effect, with PAMP injected mice traveling statistically similar distances (in cms) as the Ringers-injected mice. This suggests that the primary driver of behavioral changes between groups is pathological tau burden, caused by the presence of the P301S transgene, rather than PAMP treatment and the downstream pro-inflammatory state that is produced.

There was a strong genotype-dependent effect on hind-limb clasp behavior, with PS19+ mice demonstrating a significantly higher composite score compared to their Non-Tg littermates. Severe progressive motor dysfunction is a hallmark of the PS19 mouse model and other mouse models of tauopathy (Roberson, 2012). Roberson and colleagues demonstrate that

PS19 mice often exhibit mild motor deficits, such as hunched posture (kyphosis), hypoactivity, and difficulty feeding or grooming by 6-8 mo. These deficits generally progress rapidly as the mouse ages, with paralyzed hind-leg function occurring in some mice by 10 mo. The hindlimb behavioral test was, therefore, utilized to determine motor function by quantifying splayed leg and foot behavior in mice that are elevated by their tail. PS19 mice scored significantly higher on a composite score of leg splaying than the Non-Tg mice, as expected. However, there was no treatment effect on hind-limb clasping (retraction of hind-limbs) within the PS19 animals, suggesting exacerbation of motor impairment was not likely to be a major cause of locomotor behavioral effects in this study.

Disclosures

There were no known conflicts of interest while conducting this study.

Author Contributions

Conceptualization and Design: Joshua Keene, Bhairavi Srinageshwar, Marcia Gordon, Julien Rossignol, Dave Morgan, Gary Dunbar.

Data Collection: Joshua Keene, Taylor Desjarlais, Brianna Jackman, Bhairavi Srinageshwar, Swathi Suresh.

Analysis: Joshua Keene, Dylan Finneran, Marcia Gordon, Dave Morgan.

Writing Manuscript Rough Draft: Joshua Keene.

Edits and Final Draft: Joshua Keene, Julien Rossignol, Dave Morgan, Gary Dunbar.

Acknowledgments

Support for this study was provided by NIH R01AG051500 to Dr. David Morgan at Michigan State University and the Neuroscience Program, College of Medicine, the John G. Kulhavi Professorship in Neuroscience, and the E. Malcolm Field and Gary Leo Dunbar Endowed Chair in Neuroscience at Central Michigan University.

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